

Solar Home Lighting

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SUBSYSTEM REPORTS

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Solar Home Lighting

TEAM 3

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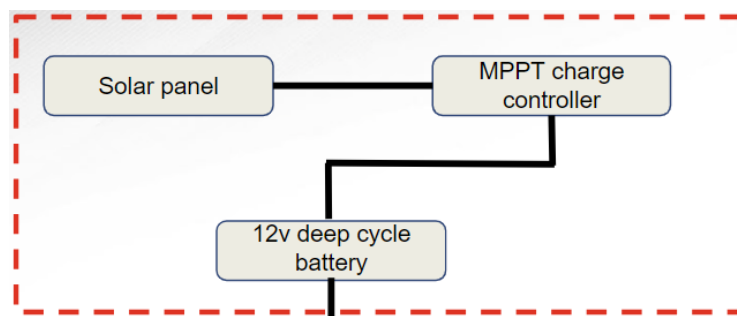
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1. Introduction

The Solar Home Lighting System is composed of several coordinated subsystems that work together to deliver reliable off-grid lighting, security monitoring, and user control. These subsystems together include the solar charge controller and battery power system, the physical control panel and power distribution hardware, the microcontroller based sensing and automation logic, and AI enabled camera with mobile application interface. Each subsystem was designed to satisfy its individual functional requirements while maintaining compatibility with the overall system as defined in the ConOps, FSR, and ICD. By validating each subsystem, it demonstrated correct operation under expected conditions

2. Charge Controller Subsystem



2.1. Subsystem Introduction

The Maximum Power Point Tracking (MPPT) charge controller is a solar energy management system designed to maximize power transfer from the solar panel to a lead-acid battery. The controller uses a microcontroller to adjust duty cycles and charging stages, ensuring safe and efficient battery charging under varying sunlight conditions. Integrated sensors measure panel voltage and current, battery voltage and current, and load current, providing real-time data that enables precise monitoring of system performance.

2.2. Subsystem Overview

The MPPT charge controller is built to maximize energy transfer from a 200 W solar panel into a 12 V, 100 Ah sealed lead-acid battery while ensuring safe charging across bulk, absorption, and float stages. At the heart of the system is the LM5176 synchronous buck-boost controller, which steps down the solar panel's operating voltage (18–22 V) to the battery's required charging levels. The ESP32 microcontroller manages system intelligence: it runs the MPPT algorithm, adjusts duty cycles through DAC/PWM outputs, and switches feedback resistor networks to set the correct charging voltage for each stage.

A suite of sensors provides real-time monitoring: ACS781LL hall-effect sensors measure panel, battery, and load currents; voltage dividers track solar and battery voltages. This data enables the ESP32 to calculate battery state of charge and charging efficiency. Safety features include over-voltage, under-voltage, and over-current, as well as a load disconnect switch to prevent battery over-discharge.

2.3. Solar Panel to Battery Buck Converter (18.8V to 14.4V - 13.6V)

2.3.1. Operation

The solar panel buck converter will regulate the input voltage from the solar panel so that the 12V lead-acid battery can safely charge. The LM5176 Buck-Boost converter is used in this design to step down the panel's nominal 18.8 V input to the battery's required charging voltages of 14.4 V (Bulk/Absorb) and 13.6 V (Float). The ESP32 microcontroller supervises the LM5176 by adjusting its feedback network and DAC/PWM control signals, allowing dynamic regulation of duty cycle to track the maximum power point under varying irradiance conditions.

Input and output filtering, compensation components, and an appropriately sized inductor are selected to support stable operation and minimize voltage deviation. Reliable performance of the LM5176 stage is essential, as excessive ripple or poor transient response can reduce charging efficiency, stress the battery, or disrupt system monitoring. The converter, therefore, serves as the fundamental power stage enabling safe, efficient energy storage and stable operation of the overall MPPT charge controller.

2.3.2. Simulation Validation

To validate the LM5176 buck converter, the charge controller circuit was designed using PSpice for TI. These simulations replicate the behavior of the LM5176 as it steps down the solar panel's input voltage to match the battery's charging requirements. The converter maintains regulation across varying input voltages and responds appropriately to control signals from the ESP32's DAC output. This confirms that the output voltage can be dynamically adjusted to support bulk, absorption, and float charging stages, even in the absence of physical hardware.

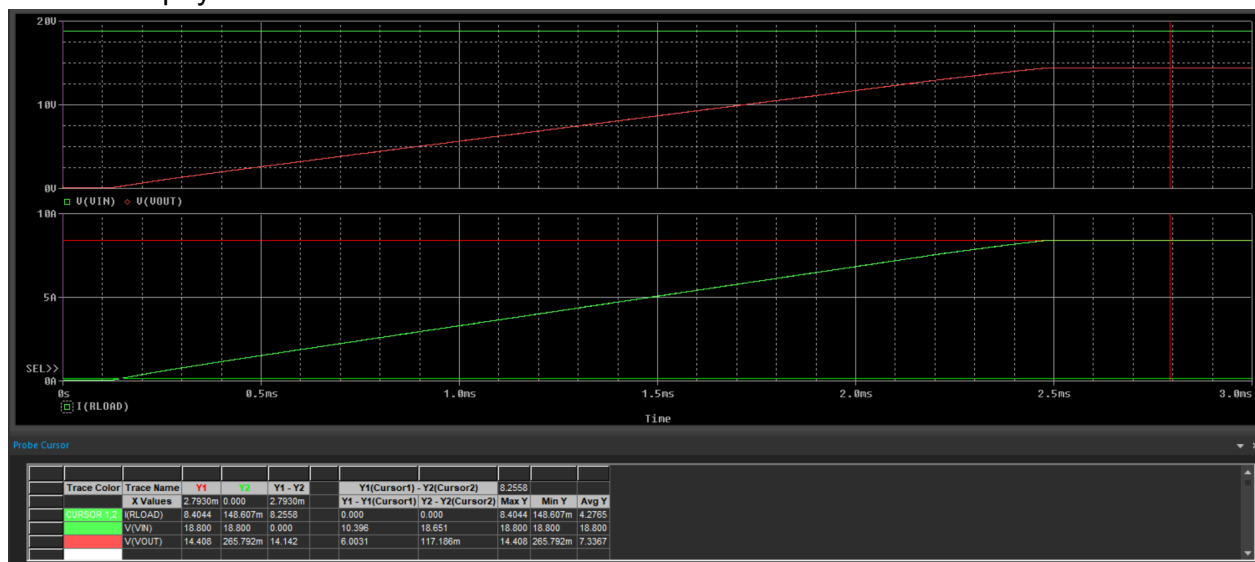


Figure 2.3.3 14.4V Buck conversion from OC Voltage

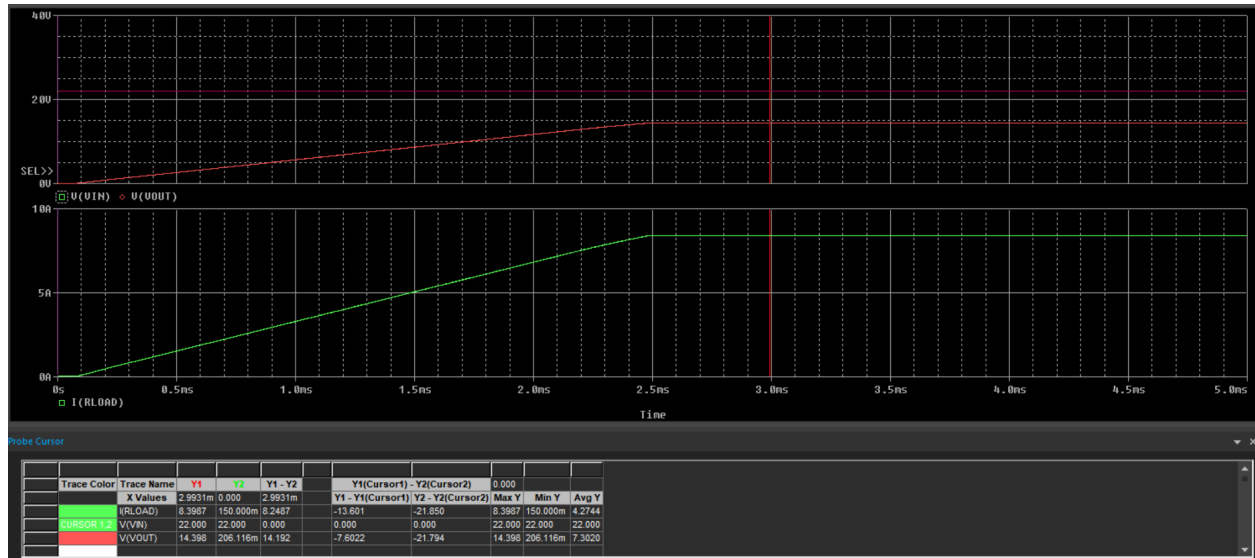


Figure 2.3.4 14.4V Buck conversion from Nominal 18.8V

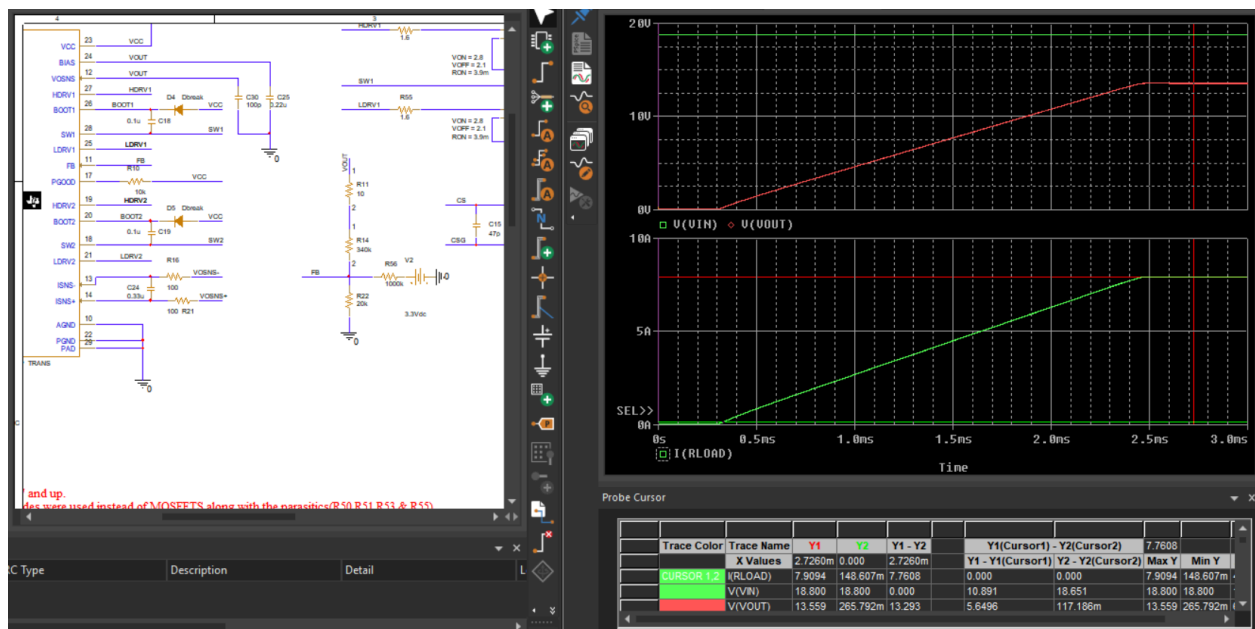


Figure 2.3.5 14.4V Buck conversion from low voltage input

2.3.2.1. 18.8V to 13.6V Step down (ESP32 DAC implementation)

A critical verification for MPPT implementation is demonstrating how the ESP32's DAC output interacts with the LM5176 feedback network to control converter duty cycle. In Figure 2.3.2.2, the DAC is modeled as a 3.3 V source connected through a 1 MΩ resistor to the LM5176 feedback pin. This configuration ensures that when the DAC reaches its maximum output, the LM5176 regulates the converter to approximately 13.6 V, corresponding to the battery's float charging stage. By scaling the DAC voltage through the resistor network, the ESP32 can dynamically bias the feedback loop, enabling precise adjustment of the converter's output voltage across different charging stages.

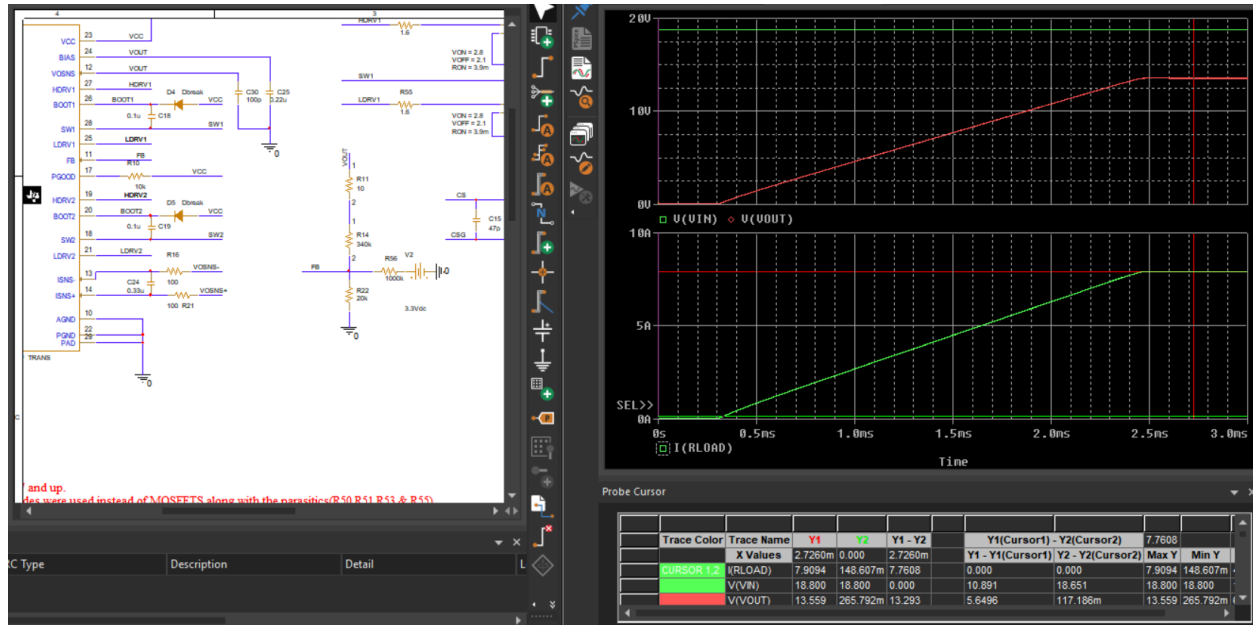


Figure 2.3.2.2 Verification of ESP32 DAC implementation

2.4. 3.3V Buck Converter

2.4.1. Operation

The 3.3 V buck converter provides the regulated logic supply required for all digital elements of the charge controller. It is implemented using the LM3150 synchronous step-down controller, which converts the 12–14.4 V from the battery bus into a stable 3.3 V output. This rail powers the ESP32 microcontroller and ACS781LL making reliable operation of the converter essential for system control and communication.

The LM3150 operates at a high switching frequency, allowing the use of compact inductors and capacitors while maintaining low output ripple and fast transient response. Input filtering ensures stable operation across the battery's voltage range, while the output filter and compensation network are tuned to support continuous conduction mode under dynamic digital loads. As the ESP32 and communication peripherals draw varying current, particularly during Wi-Fi transmission, UART logging, and Bluetooth activity, the converter must maintain regulation across rapidly changing load conditions.

2.4.2. Simulation Validation

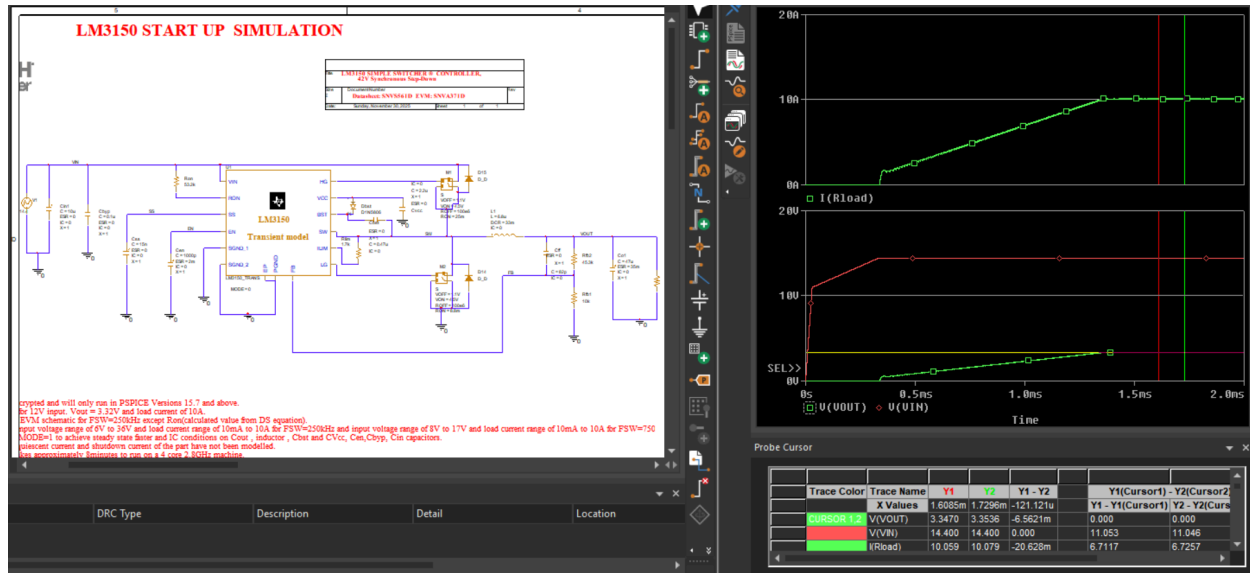


Figure 2.1.3 3.3V LM3150 Buck converter during Bulk/Absorption Stage

2.5. ESP32 Perturb & Observe (P&O) Program

2.5.1. Operation

The ESP32 microcontroller coordinates power conversion, sensing, protection, and communication. It executes the maximum power point tracking (MPPT) algorithm, dynamically adjusting the LM5176 buck converter through DAC and PWM outputs to optimize energy transfer from the solar panel. The ESP32 continuously acquires data from integrated sensors, including voltage dividers for panel and battery monitoring, and hall-current sensors for current measurement. With this information, the ESP32 will calculate the efficiency of the system and battery State of Charge (SOC). The ESP32 reads solar panel and battery voltages/currents, calculates power and efficiency, tracks battery state of charge (SOC), and drives the LM5176 buck converter via its DAC output to regulate charging. It uses a state machine to switch between charging stages

2.5.2. Program Validation

To validate program operation, test cases were developed to emulate the behavior of the solar panel and battery, ensuring that the ESP32 correctly drives the DAC output. Figure 2.5.3 illustrates the relationship between battery state of charge (SOC) and DAC value. As SOC increases, the DAC adjusts to regulate the LM5176 feedback network accordingly. During the Bulk stage, the battery draws high current, and maximum power point tracking (MPPT) is most effective for rapid charging. Once the battery approaches ~80% SOC or current demand decreases significantly, MPPT provides diminishing returns, and the converter transitions to the Absorption stage, holding the battery at 14.4 V until current tapers. At full capacity, the system enters the Float stage, where the DAC sets a lower

target voltage (13.6 V) to maintain charge while preventing overcharging and minimizing battery stress.

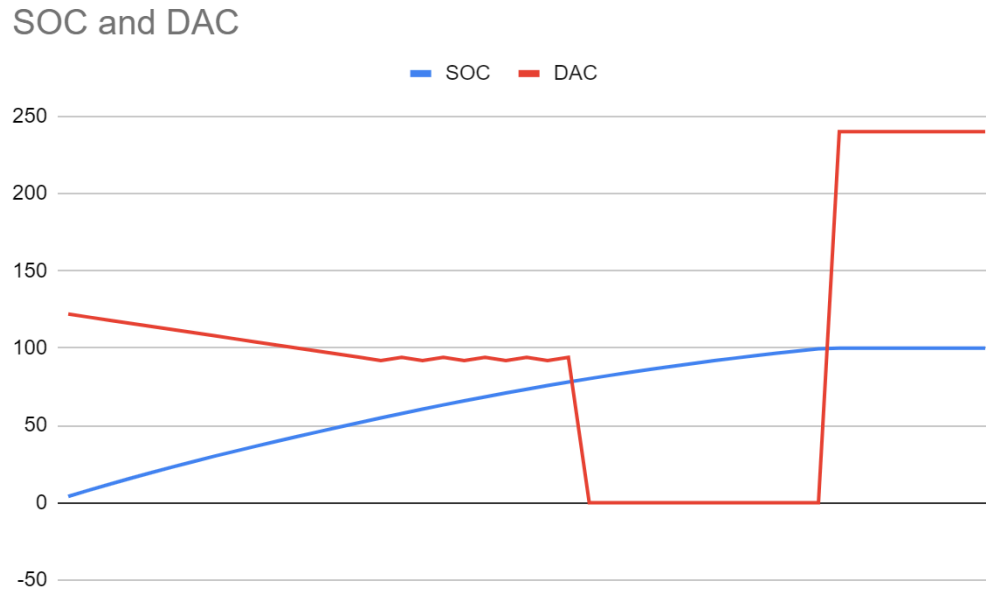


Figure 2.5.3 ESP32 adjusting DAC according to Battery SOC

Figure 2.5.4 illustrates the implementation of maximum power point tracking (MPPT) during the Bulk stage. The ESP32 monitors system efficiency while perturbing the DAC output to the LM5176 feedback pin using the Perturb and Observe (P&O) method. The DAC value is adjusted incrementally. If a change results in higher efficiency, the controller continues in the same direction, if efficiency decreases, the algorithm reverses direction and steps the DAC opposite. Through this iterative process, the ESP32 converges toward the operating point that maximizes panel power transfer to the battery.

DAC vs. efficiency for Bulk

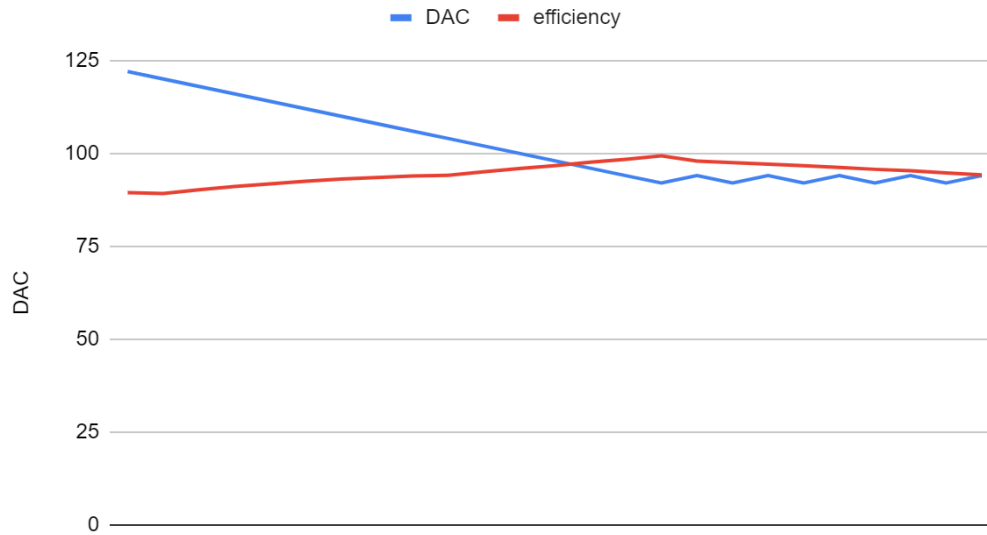


Figure 2.5.4 ESP32 P&O according to efficiency of system

2.6. Subsystem Conclusion

As the solar input and battery load fluctuate, the LM5176 converter must maintain regulation across rapidly changing operating points, ensuring continuous conduction and efficient energy transfer. Real-time information from the sensor components allows efficient regulation of charging stages while enforcing safety cutoffs for over-voltage, under-voltage, over-current, and deep discharge. By integrating control logic, sensing, protection, and communication, the ESP32 transforms the LM5176 synchronous buck-boost converter into an adaptive solar energy management system.

3. Physical Controls Subsystem

3.1. Subsystem Introduction

The Physical Controls Subsystem provides the local interface, power regulation, and lighting-control capability required for the Smart Solar Residential Lighting and Security System. It enables the homeowner to manually operate lighting zones through a touchscreen display while also supporting the microcontroller logic that executes user commands and routes control signals to the relay hardware. In addition to handling all user interaction, the subsystem delivers stable low-voltage power to the system's electronics through dedicated DC-DC conversion stages.

This subsystem forms the foundation for local operation of the overall system by supplying regulated power, processing interface inputs, and actuating the 12 V lighting loads. The following sections describe its electrical design, hardware operation, and validation testing.

3.2. Subsystem Overview

The Physical Controls Subsystem begins with a 10.8-14.2 V rail which is distributed to two internal domains: the relay hardware, which requires a stable 12 V supply for coil actuation, and a dedicated 3.3 V buck converter that generates the logic-level rail for all digital electronics. The regulated 3.3 V output powers both ESP32-S3 microcontrollers as well as the 2.8-inch ILI9341 TFT display and FT6206 capacitive touch controller. These elements make up the digital control layer of the subsystem. The touchscreen ESP32 functions as the human-machine interface, rendering the graphical control panel and interpreting user input. When the user selects a lighting function, the touchscreen microcontroller converts this interaction into a structured control command and transmits it over a dedicated UART link to the relay-control ESP32. The relay-control ESP32 receives these commands and responds by actuating the appropriate relay channels on the five-channel relay module. Each relay is driven through an isolated transistor interface, allowing the 3.3 V logic domain to safely control 12 V lighting loads. The relay controller returns acknowledgment messages back to the touchscreen ESP32 to maintain consistent system state awareness.

Through this hierarchy—input voltage distribution from 10.8 to 14.2 V, regulated power conversion, coordinated microcontroller communication, and relay-based switching—the subsystem delivers reliable and responsive lighting control under all expected operating conditions. The following sections describe the detailed operation and validation of each major component: the 3.3 V converter, the 12 V converter, and the ESP32 touchscreen and relay-control system.

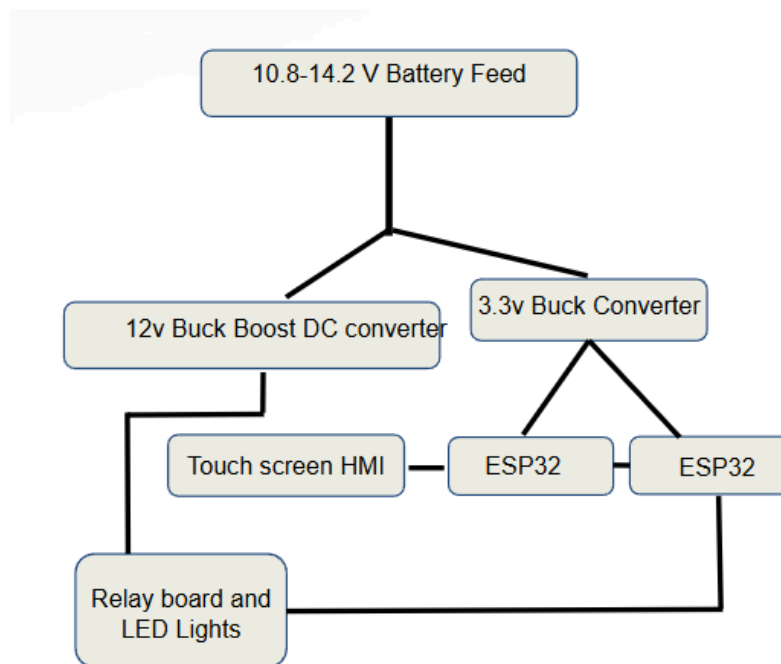


Figure 3.2.1. Physical Controls System Overview

3.3. 3.3V Buck Converter

To confirm that the 3.3 V converter met expected performance, a series of electrical tests were conducted under controlled bench conditions. These tests evaluated the converter's ability to maintain stable voltage output across varying input voltages and load currents.

3.3.1. Operation

The 3.3 V buck converter provides the regulated logic supply for all digital elements of the Physical Controls Subsystem. It is implemented using the LT8610AC synchronous step-down regulator, which converts the subsystem's incoming 10.8–14.2 V rail to a stable 3.3 V output. The converter operates at a high switching frequency, allowing the use of compact inductors and capacitors while maintaining low ripple and fast transient response. This 3.3 V rail powers both ESP32-S3 microcontrollers as well as the 2.8-inch ILI9341 TFT display and FT6206 capacitive touch controller. As these devices draw dynamically varying current—particularly during Wi-Fi transmission, display refresh operations, and UART communication—the converter must maintain regulation across rapidly changing load conditions. The LT8610AC design includes appropriate input and output filtering, compensation components, and an inductor selected to support stable continuous conduction across the subsystem's operational range. Reliable operation of the buck converter is essential, as voltage deviation or excessive ripple can produce ESP32 brownout resets or display instability. The converter therefore serves as the fundamental power stage enabling stable digital control and user interaction.

3.3.2. Validation

Validation of the 3.3 V converter was performed to ensure proper electrical behavior across expected operating conditions. Testing focused on confirming stable output regulation, acceptable ripple performance, and consistent efficiency.

3.3.2.1. Line Regulation

Line regulation was evaluated by sweeping the input voltage from 10.8 V to 14.2 V under two load conditions: no load and a high load of 1.75 A. The results for both cases are shown in Figures 3.3.2.1 and 3.3.2.2.

Under no-load conditions ($I_{out} = 0$ A), the converter output varied by less than 0.1% across the entire input range. The curve remained nearly flat, indicating that changes in input voltage had minimal effect on the 3.3 V output. This behavior confirms that the LT8610AC maintained strong closed-loop control even in a low-current state.

When tested at 1.75 A, representing a near-maximum subsystem load, the output deviation increased slightly, ranging from approximately -0.55% to -0.77% . Although the magnitude of variation was greater than in the no-load condition, the response remained smooth and well within acceptable tolerance for ESP32 operation. No signs of instability, oscillation, or dropout were observed during the sweep.

Overall, the converter demonstrated stable performance across the full input voltage range. The measured deviations are consistent with expected behavior of a buck converter operating under varying duty cycles, and the results verify that the 3.3 V rail remains properly regulated for all expected operating conditions of the subsystem.

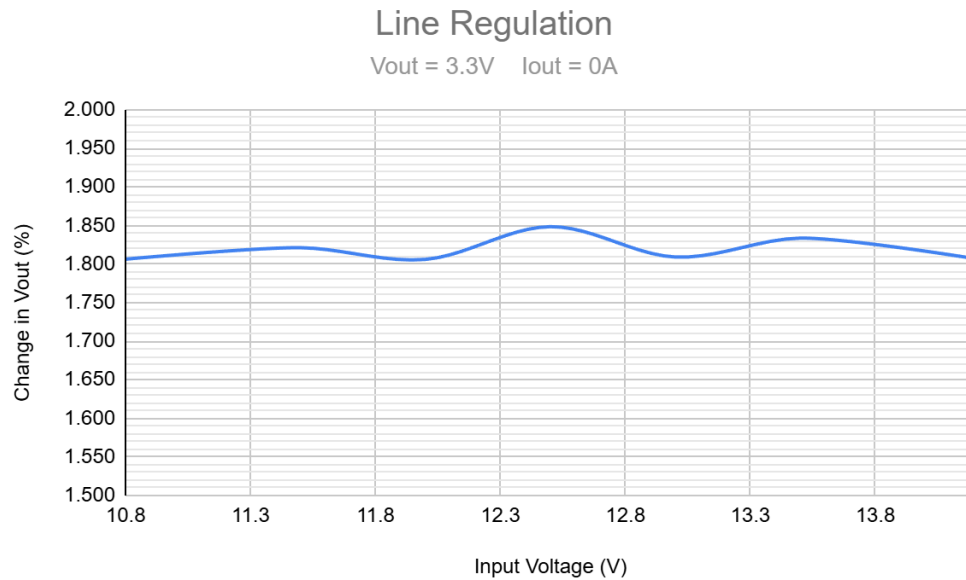


Figure 3.3.2.1. Line Regulation Iout = 0A

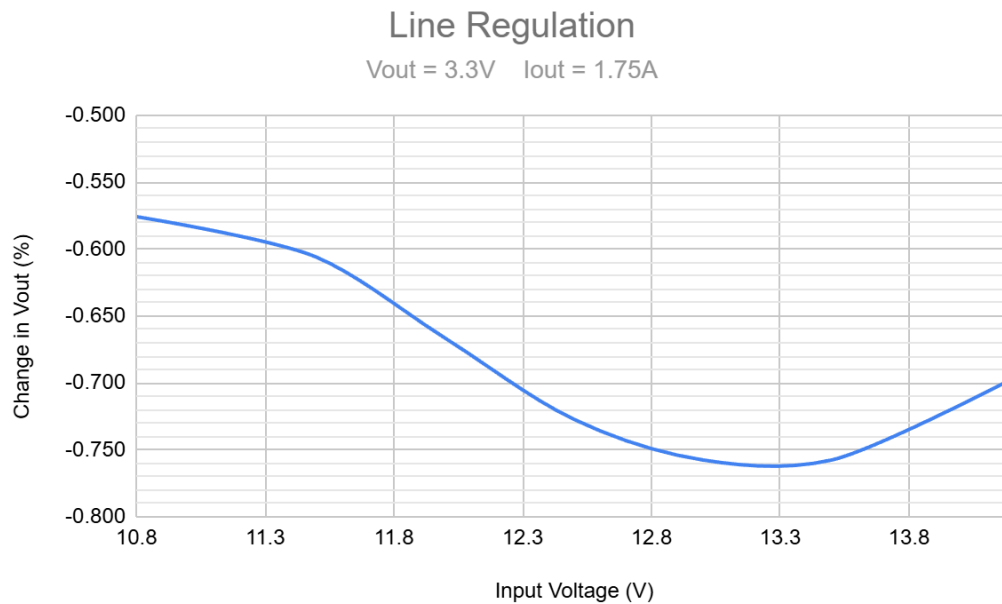


Figure 3.3.2.2. Load Regulation Iout = 1.75A

3.3.2.3. Load Regulation

Load regulation was evaluated by increasing the output current from 0 A to 2.0 A, measuring the percent change in output voltage at input voltages of 10.8 V, 12 V, and 14.2 V. The results are presented in Figure 3.3.2.3.

Across all three input voltages, the converter showed a smooth and controlled decrease in output voltage as the load increased. From 0 A up to approximately 1.7 A, the change in V_{out} remained within about -0.5% to -2.5%, and the curves for all input voltages were closely aligned. This indicates that the converter maintained stable output regulation across the majority of the load range, with performance largely unaffected by variations in V_{in} .

As the load approached the upper limit of the test, the converter exhibited a more rapid voltage drop. At the full 2.0 A load, V_{out} deviation increased to approximately -5.5% to -6.0% depending on input voltage. This behavior is expected when operating near the converter's maximum current capability, where conduction losses, switch resistance, and inductor saturation effects become more significant.

Overall, the converter maintained acceptable regulation across its entire tested range up to 2.0 A. Output degradation occurred only at the extreme high-load condition, confirming that the 3.3 V rail remains suitable for all expected subsystem operating currents.

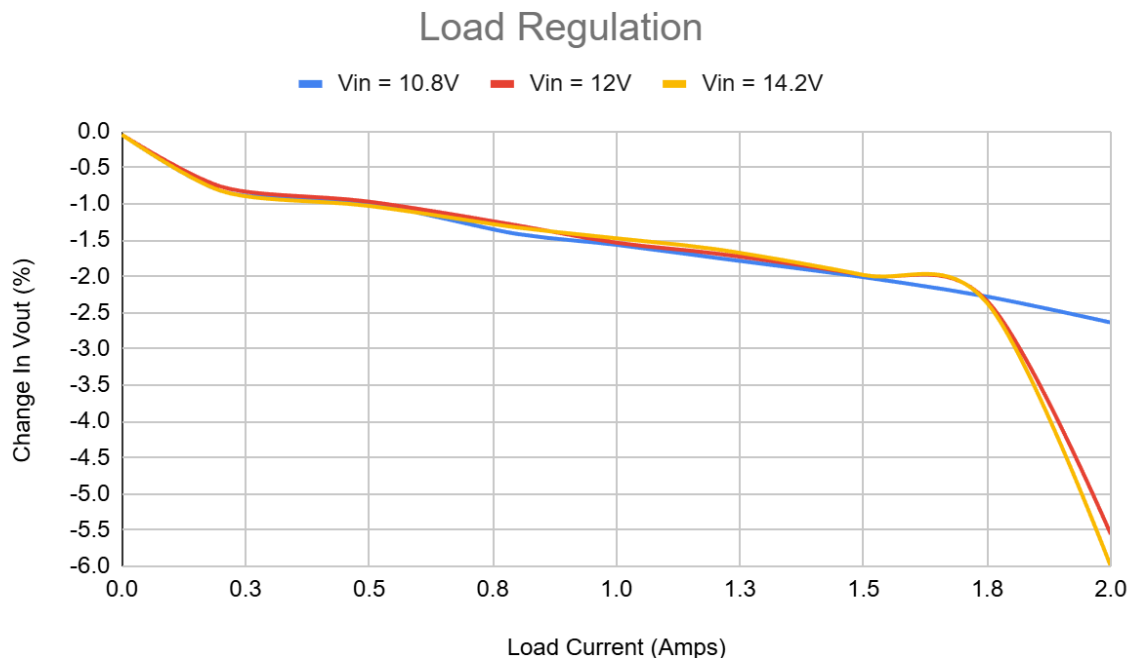


Figure 3.3.2.3. Load Regulation

3.3.2.4. Efficiency

Efficiency was measured across load currents from 0.2 A to 2.0 A for input voltages of 10.8 V, 12 V, and 14.2 V. The results are shown in Figure 3.3.2.4.

For all three input conditions, efficiency increased rapidly as load current rose from 0.2 A to approximately 0.7 A, reaching a peak of about 92-93%. This operating region corresponds to the converter's optimal balance of switching and conduction losses. Beyond 0.7 A, efficiency gradually decreased with increasing load, falling to approximately 88-89% at 1.5 A.

As the converter approached its maximum tested load of 2.0 A, efficiency dropped more noticeably. At this point, efficiency reached roughly 82-85%, depending on the input voltage, with the lowest efficiency occurring at $V_{in} = 14.2$ V. This behavior is consistent with increased conduction losses and the higher duty cycles required to maintain output regulation under heavy load.

Across the entire tested range, efficiency curves for all input voltages were closely aligned, indicating that V_{in} had only a minor effect on performance. The converter demonstrated high efficiency in the load region most representative of normal subsystem operation (0.3–1.2 A), confirming suitability for continuous powering of the ESP32 microcontrollers and display hardware.

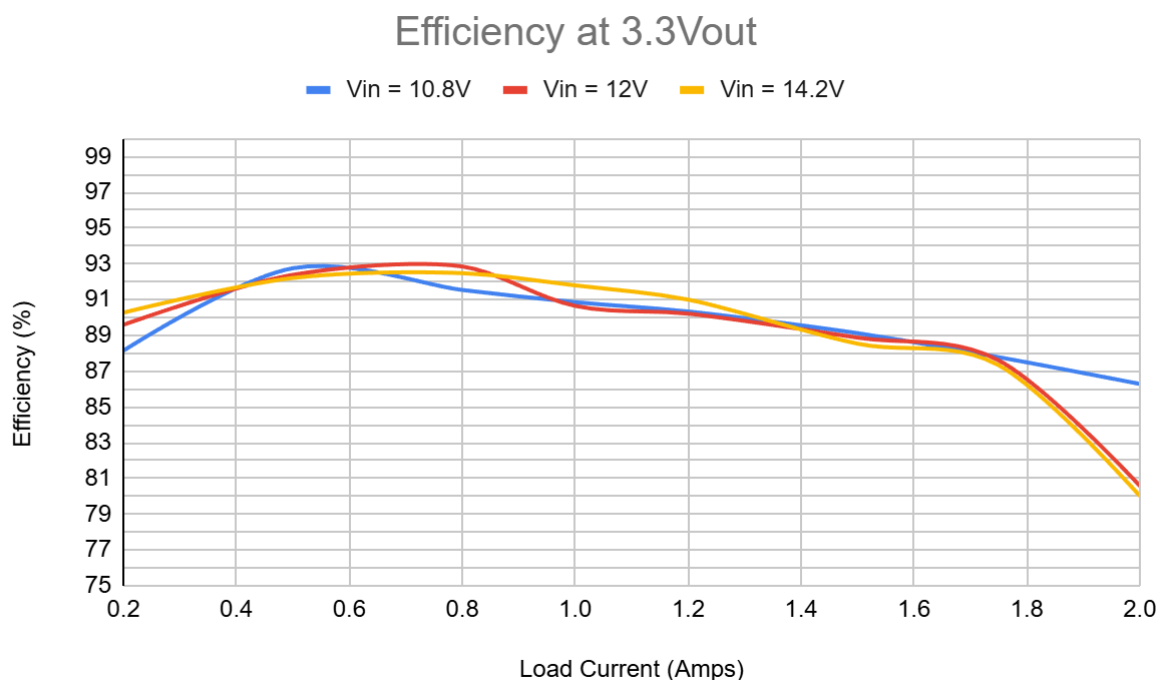


Figure 3.3.2.4. Efficiency at 3.3V

3.3.2.5. Output Voltage Ripple

Ripple measurements were taken under two conditions: no load and a nominal operating load of 0.8 A. At no load, the converter produced approximately 71 mV peak-to-peak ripple on the 3.3 V output (Figure 3.3.2.5.1). This value aligns with expectations for a high-frequency buck converter operating with minimal inductor current and reduced energy storage.

Under nominal load conditions ($I_{out} = 0.8 \text{ A}$), ripple increased slightly to approximately 90 mV peak-to-peak (Figure 3.3.2.5.2). This rise in ripple is consistent with higher inductor current ripple at increased load and the converter operating closer to its continuous conduction region. The magnitude of ripple remained stable, with no signs of oscillation, discontinuities, or switching irregularities.

Across both test conditions, the observed ripple remained well within acceptable limits for powering ESP32 microcontrollers and the TFT display, neither of which showed instability or brownout behavior during operation. The results confirm that the output filtering network is adequate for the subsystem's digital load requirements.

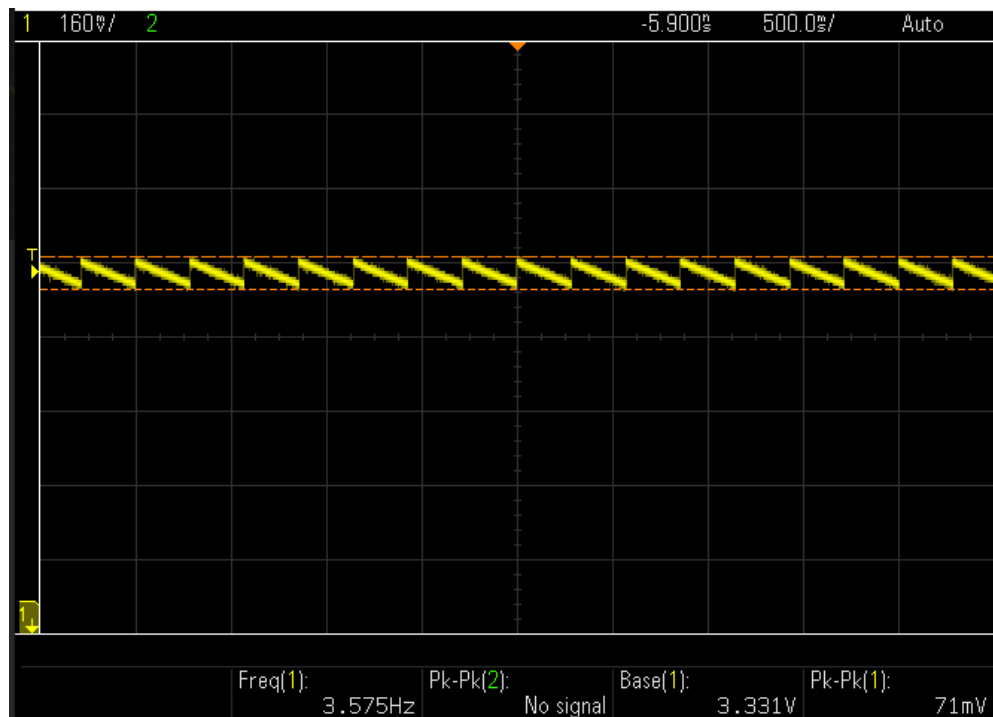


Figure 3.3.2.5.1. No load

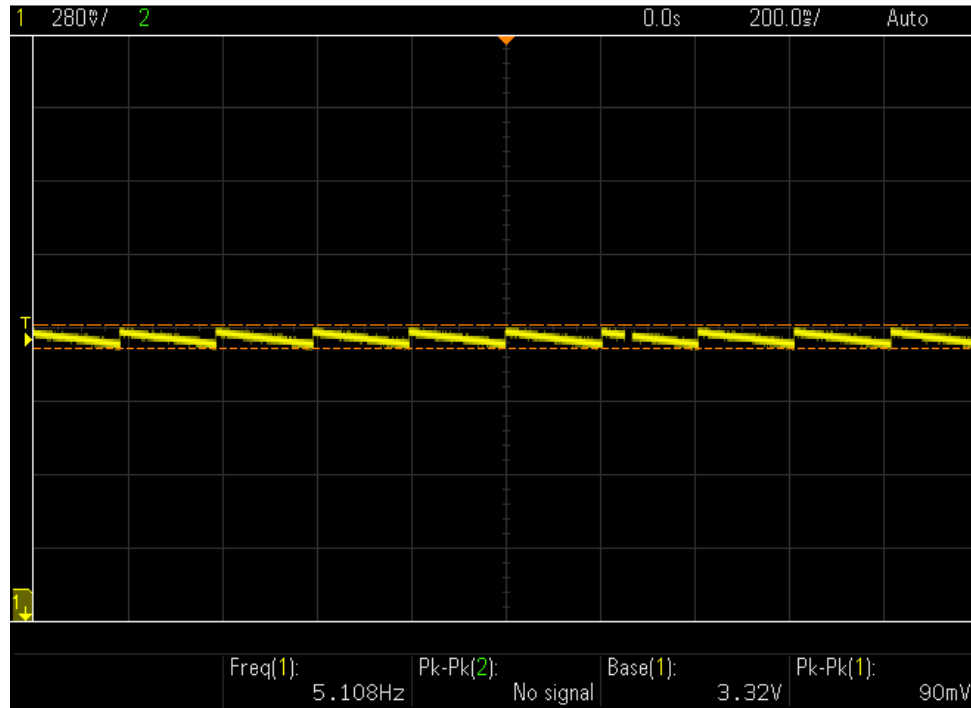


Figure 3.3.2.5.2. Nominal Load $I_{out} = 0.8$ Amps

3.3.2.6. Percent Error

Percent error was calculated across a range of input voltages and load conditions to quantify the deviation of the converter's output from the nominal 3.3 V setpoint. The results are summarized in Table 3.3.2.6.

Under no-load conditions, output voltage measured approximately 3.357-3.359 V, corresponding to a percent error of 1.73-1.79%. This slight positive deviation is expected due to minimal load current and the converter operating in a low-duty, discontinuous conduction region.

At nominal load (0.8 A), accuracy improved significantly, with percent error reduced to 0.36–0.48% across the input voltage range. This reflects the converter transitioning into a more stable conduction mode, where switching behavior and inductor current are better regulated.

At normal maximum load (1.75 A), the converter displayed small negative deviations, ranging from -0.52% to -0.61%, indicating a slight drop in output voltage under higher current demand. These values remain well within acceptable limits for the ESP32 and associated logic circuitry.

At the absolute maximum tested load (2.0 A), percent error increased to -0.88% to -4.15%, with the largest deviation occurring at $V_{in} = 14.2$ V. This increased drop corresponds to the converter operating near its upper current limit, where conduction losses and switch resistance become more significant.

Across the full range of operating conditions, the percent error remained within levels that support stable microcontroller and display operation. Only the highest load case showed notable deviation, which aligns with the design's expected maximum load boundary.

Vin	Iin	Iout	Vout	State	%error
10.8	0	0	3.358	No Load	1.76%
12	0	0	3.359	No Load	1.79%
14.2	0	0	3.357	No Load	1.73%
10.8	0.268	0.8	3.312	Nominal Load	0.36%
12	0.242	0.8	3.313	Nominal Load	0.39%
14.2	0.202	0.8	3.316	Nominal Load	0.48%
10.8	0.606	1.75	3.283	Normal Max	-0.52%
12	0.546	1.75	3.28	Normal Max	-0.61%
14.2	0.463	1.75	3.281	Normal Max	-0.58%
10.8	0.703	2	3.271	Absolute Max	-0.88%
12	0.647	2	3.182	Absolute Max	-3.58%
14.2	0.549	2	3.163	Absolute Max	-4.15%

Table 3.3.2.6. Percent Error Results

3.4. 12V Buck-Boost Converter

3.4.1. Operation

The 12 V rail is generated using the LTC3119 buck-boost converter, selected for its ability to maintain a regulated 12 V output from the system's variable input range of 10.8-14.2 V. The converter was designed to support a normal operating load of 0.7-1.0 A, with a continuous output capability of 2 A and a peak load tolerance of up to 3 A for short-duration relay actuation or transient events.

The buck-boost topology is intended to ensure uninterrupted light functionality during battery discharge, seamlessly transitioning between buck and boost operation as required. The design follows the LTC3119 datasheet recommendations for inductor current rating, compensation network, and input/output capacitor values. Under correct implementation, the converter should provide sufficient power for all lighting loads.

However, during system testing, the converter operated reliably only up to approximately 0.85 A, significantly below the intended 2-3 A capability. Beyond this point, the converter failed to maintain regulation and output voltage collapsed. This load limitation directly affected the ability to validate the converter at its full design rating and prompted further investigation into potential hardware faults.

3.4.2. Partial Validation

Validation testing assessed converter performance across line and load conditions within the functional range available on the assembled hardware.

3.4.2.1. Line Regulation

Figures 3.4.2.1 and 3.4.2.2 show the measured line regulation performance of the 12 V buck-boost converter at 0 A and 0.8 A, respectively. Testing was conducted over the full operating input range of 10.8 V to 14.2 V, reflecting the expected voltage swing of a typical residential battery system.

At 0 A load, the converter output deviated between approximately -0.30% and 0%, indicating relatively tight regulation when the converter is lightly loaded. The output remains close to nominal across all input voltages, with no significant drift or instability as V_{IN} increases. Minor variations appear between 12.3 V and 13.3 V, but remain well below 0.3%, showing good unloaded regulation behavior.

At 0.8 A, the highest current the hardware could sustain, the output deviation increases slightly, ranging from -1.0% to -0.7%, but the overall trend remains consistent across the input sweep. The voltage droop deepens at the lower end of the V_{IN} range and improves modestly around mid-range input levels before rising slightly near 14 V. No oscillatory behavior or step changes were present, confirming stable operation within the converter's reduced functional load capability.

Overall, the line regulation results demonstrate that the converter maintains predictable behavior across varying input voltages for both no-load and functional-load conditions. Although the converter could not reach its intended operating range of 2 A nominal, the partial data indicates that within the 0-0.8 A region the converter remains stable and exhibits regulation performance that follows expected buck-boost characteristics.

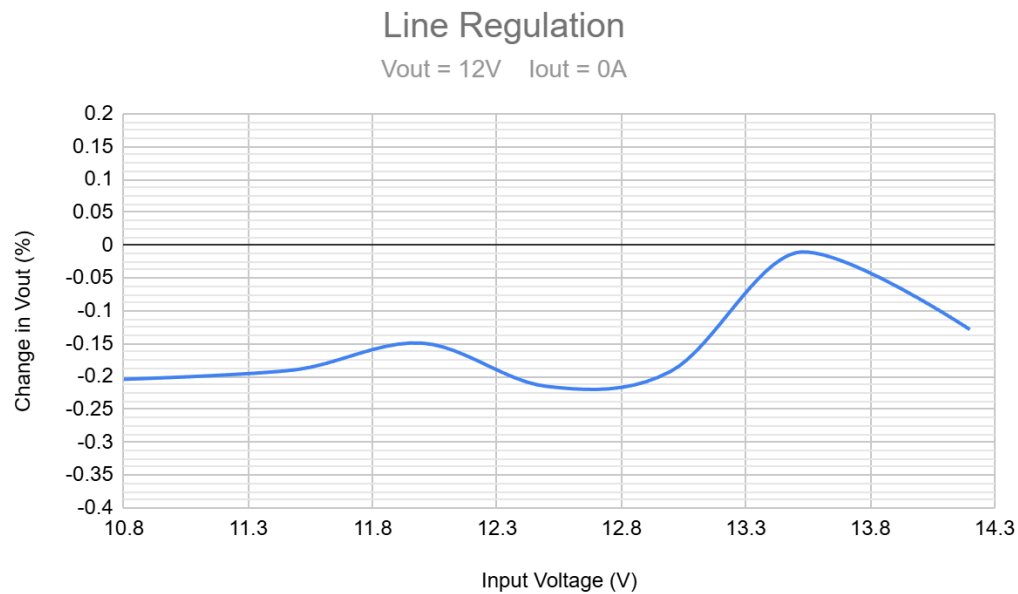


Figure 3.4.2.1. Line Regulation $I_{out} = 0A$

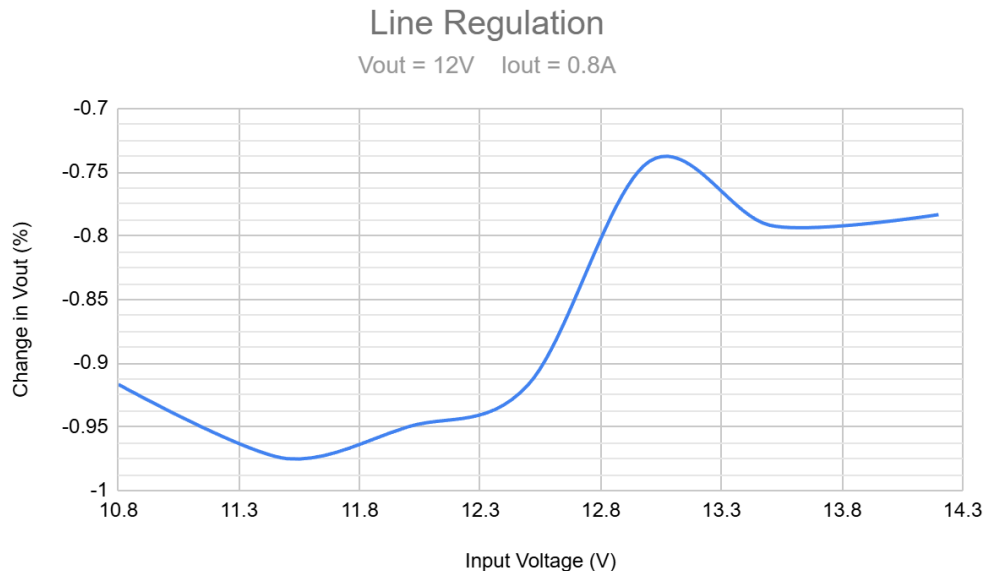


Figure 3.4.2.2. Load Regulation $I_{out} = 1.75A$

3.4.2.3. Load Regulation

Figure 3.4.2.3 illustrates the load-regulation performance of the 12 V buck-boost converter for the three tested input voltages (10.8 V, 12 V, and 14.2 V) across the converter's functional load range of 0.1 A to 0.8 A. Although the converter was designed to support up to 2A continuous output, hardware limitations restricted successful operation to approximately 0.8 A. Within this region, the converter remained stable and produced consistent output-voltage behavior.

Across all three curves, the output voltage shows a gradual negative deviation as load current increases. For $V_{IN} = 10.8$ V, the converter demonstrates the best overall regulation, with deviations beginning near -0.25% at 0.1 A and decreasing to approximately -0.55% at 0.8 A. The slope is gradual, indicating that the converter maintains reasonable voltage control despite approaching the edge of its functional current capability.

For $V_{IN} = 12$ V, regulation remains relatively flat at lighter loads, with deviations clustered near -0.30% up to around 0.3 A. Beyond this point, the output droop becomes more pronounced, reaching approximately -0.60% at 0.8 A. This trend indicates that the converter handles moderate load conditions adequately but begins to show increased voltage sag as the current approaches the functional limit.

For $V_{IN} = 14.2$ V, the converter exhibits the greatest deviation at light load, beginning around -0.40% . The regulation briefly improves near 0.25 A before steadily declining with increasing load, ultimately reaching approximately -0.70% at 0.8 A. This behavior suggests reduced efficiency at higher input voltage, consistent with the converter's elevated switching and conduction losses at lighter load conditions.

Overall, the load-regulation results indicate that the converter maintains predictable and stable performance within its reduced operating range. Although output droop increases with load, no instability or oscillatory behavior was observed, and the trends remain consistent with expected buck-boost converter characteristics operating below their nominal current capability.

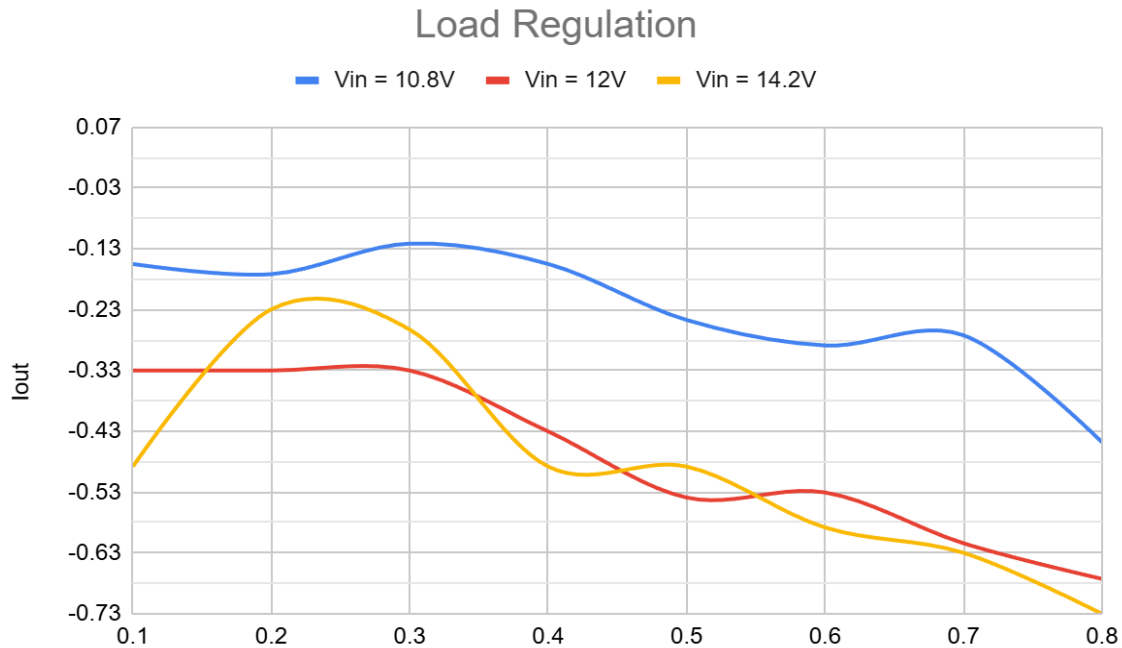


Figure 3.4.2.3. Load Regulation

3.4.2.4. Efficiency

Figure 3.4.2.4 shows the measured efficiency of the 12 V buck-boost converter across load currents from 0.1 A to 0.8 A for three input voltages (10.8 V, 12 V, and 14.2 V). Although the converter was originally designed for higher load capability, only data up to 0.8 A is shown due to the converter's functional limitation above this range. Within the operating region, efficiency follows a consistent and predictable trend.

For $V_{IN} = 10.8 V$, efficiency starts at approximately 89% at 0.1 A and increases steadily with load, reaching a peak of nearly 96% around 0.6 A, after which it tapers slightly but remains above 93% at 0.8 A. This curve represents the best overall performance, as lower input voltage typically reduces switching losses in buck-boost converters operating near unity conversion ratio. For $V_{IN} = 12 V$, the converter demonstrates a slower rise in efficiency, beginning near 85% at 0.1 A and climbing to roughly 93% at 0.5–0.6 A. Beyond this region, efficiency decreases marginally due to increasing conduction losses and the converter approaching its functional current limit. Overall, the performance remains within an acceptable range for a converter operating in buck-boost mode. For $V_{IN} = 14.2 V$, efficiency is the lowest at light load (approximately 76% at 0.1 A), likely due to increased switching and gate-drive losses at higher input voltage. Efficiency improves significantly with load, peaking around 94–95% near 0.6 A, and stabilizes above 93% as the converter approaches

0.8 A. This trend aligns with expected behavior: as load increases, the converter operates in a region where switching losses are proportionally less dominant.

Across all input conditions, the converter demonstrates strong mid-load efficiency performance, with peak efficiency values above 93% and in some cases approaching 96%. These results indicate that, despite the reduced maximum-load capability, the converter operates efficiently within its functional range and is well-suited for the system's typical current demands (0.7–1.0 A).

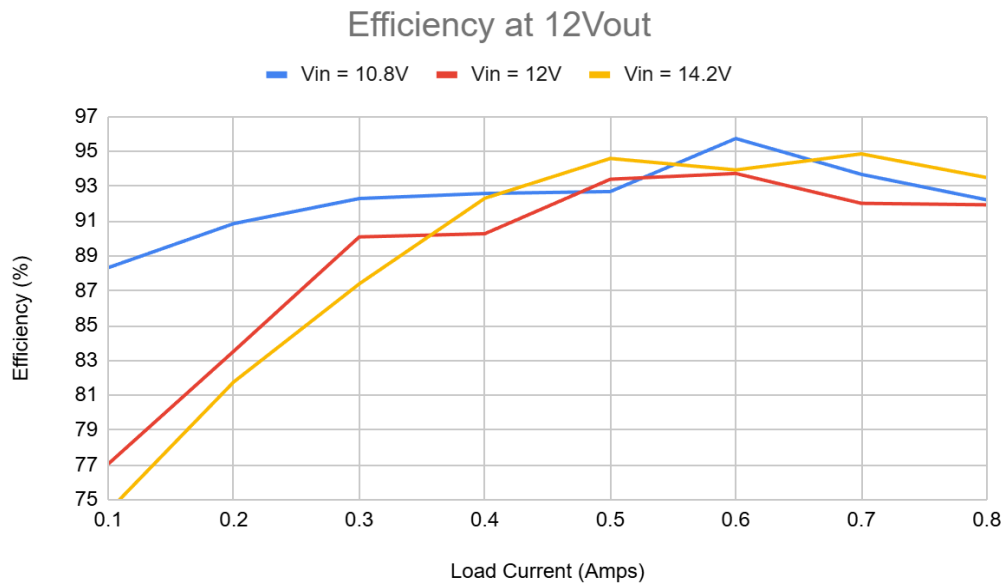


Figure 3.4.2.4. Efficiency at 3.3V

3.4.2.5. Output Voltage Ripple

Ripple measurements for the 12 V converter are shown in Figures 3.4.2.5.1 and 3.4.2.5.2. At no load, the output exhibited approximately 181 mV peak-to-peak ripple.

Under nominal load (0.8 A) the highest current the converter could sustain before failing the ripple increased to approximately 325 mV peak-to-peak.

Both values are significantly higher than expected for this converter architecture and exceed typical ripple limits for a stable buck-boost stage. The elevated ripple is consistent with the hardware fault identified during testing, in which the converter was unable to deliver current beyond ~0.85 A. This degraded operating condition likely caused improper switching behavior and reduced filtering effectiveness, leading to the unusually large ripple amplitudes.

Because the unit did not reach its intended operating region, these ripple results represent the behavior of a partially failed converter rather than the expected performance of the design.

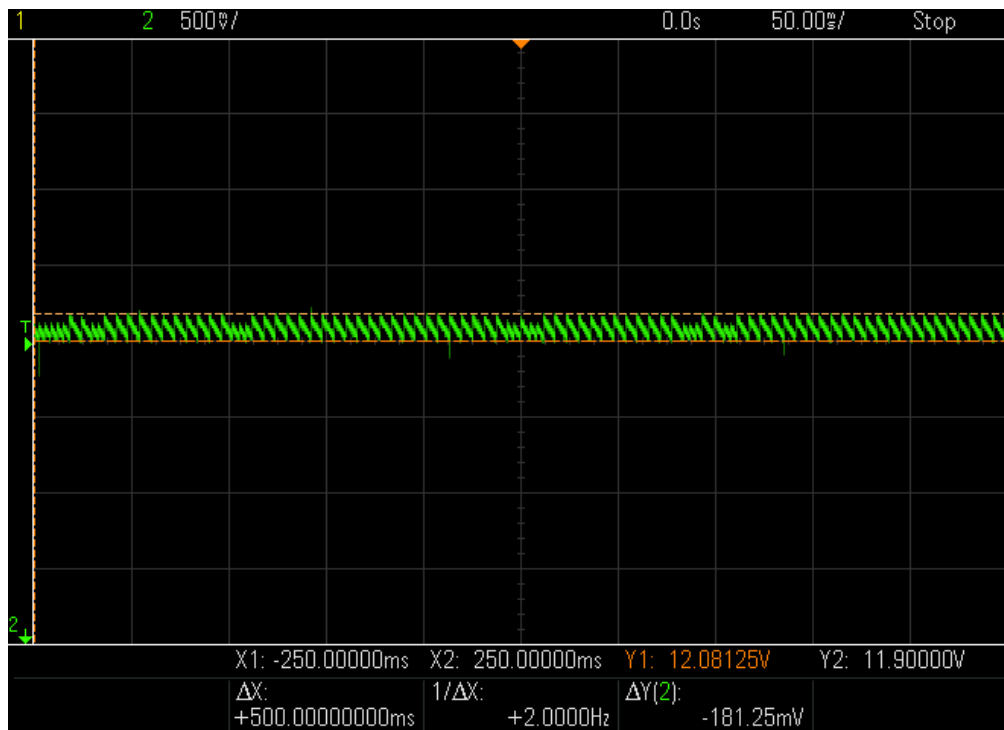


Figure 3.4.2.5.1. No load

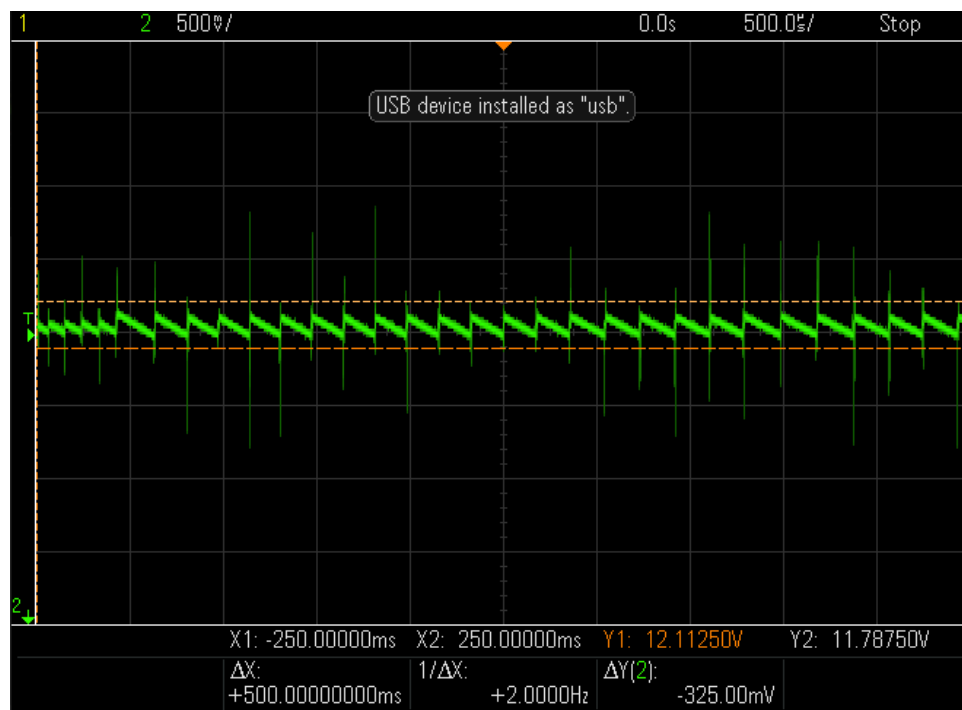


Figure 3.4.2.5.2. Nominal Load $I_{out} = 0.8$ Amps

3.4.2.6. Percent Error

Table 3.4.2.6 summarizes the percent error of the 12 V converter across all measured operating points.

Under no-load conditions, the output remained within -0.13% to -0.25%, demonstrating stable voltage regulation when the converter is lightly loaded.

At the single-light and dual-light load levels (approximately 0.45 A and 0.8 A), the percent error ranged from -0.70% to -0.96%, which is still within an acceptable tolerance for a 12 V regulation stage. The gradual increase in negative error reflects the additional voltage drop as the converter is brought closer to its functional limit.

The final operating region—labeled “Limit”—captures the highest load currents the converter was physically able to sustain: 0.855–1.0 A, depending on input voltage. At these points, the percent error increased to -1.0% to -1.10%, and the converter was unable to provide additional current. This behavior confirms the earlier observation that the hardware fault prevented the converter from reaching its intended design range (2 A nominal, 3 A peak). The percent-error values in this region represent operation at the edge of failure rather than normal performance.

Because the converter’s current ceiling was reached prematurely, these percent-error results do not reflect the expected accuracy of the full design but instead illustrate the degraded behavior of the partially failed unit.

Vin	lin	Iout	Vout	State	%error
10.8	0	0	11.97	No Load	-0.25%
12	0	0	11.984	No Load	-0.13%
14.2	0	0	11.983	No Load	-0.14%
10.8	0.526	0.45	11.912	Single Light Load	-0.73%
12	0.481	0.45	11.907	Single Light Load	-0.78%
14.2	0.405	0.45	11.916	Single Light Load	-0.70%
10.8	0.935	0.8	11.891	Dual Light Load	-0.91%
12	0.859	0.8	11.885	Dual Light Load	-0.96%
14.2	0.708	0.8	11.903	Dual Light Load	-0.81%
10.8	0.979	0.855	11.877	Limit	-1.02%
12	0.898	0.865	11.881	Limit	-0.99%
14.2	0.884	1	11.868	Limit	-1.10%

Table 3.4.2.6. Percent Error Results

3.4.1. Failure Root Cause Analysis

During validation, the 12 V DC-DC converter operated correctly up to approximately 0.8 A of load current. Above ~0.9–0.93 A, the converter consistently entered a collapse-and-restart failure mode in which VOUT dropped to only a few millivolts, momentarily recovered, and then failed again. VIN remained stable during all events, and the measured IC temperature during testing was low (~94 °F), indicating the failure was not caused by normal operating thermal stress. Instead, the failure is attributed to a combination of PCB grounding/layout issues and possible internal degradation of the LTC3119 IC caused by overheating during assembly.

Figure 3.4.1.1 shows the VCC/VC node under correct operation at ~0.8 A. The waveform remains relatively stable, with small switching-related perturbations as expected for a functional compensation loop.

However, as shown in Figure 3.4.1.2, once the converter reaches approximately 0.93 A, the VCC/VC waveform becomes a distorted, noisy signal that resembles a corrupted version of the SW1 switching node. Instead of a clean bias supply, the IC's internal control voltage is overtaken by switching noise. When the effective VCC drops or becomes unstable, the LTC3119's internal protection circuitry disables switching. With switching halted, VCC temporarily rises again, causing the chip to attempt a restart—producing the cyclical collapse observed during testing.

This distorted control-voltage behavior directly indicates a loss of control-loop stability and instability in the device's internal bias generator, which strongly supports a grounding-related failure mechanism.

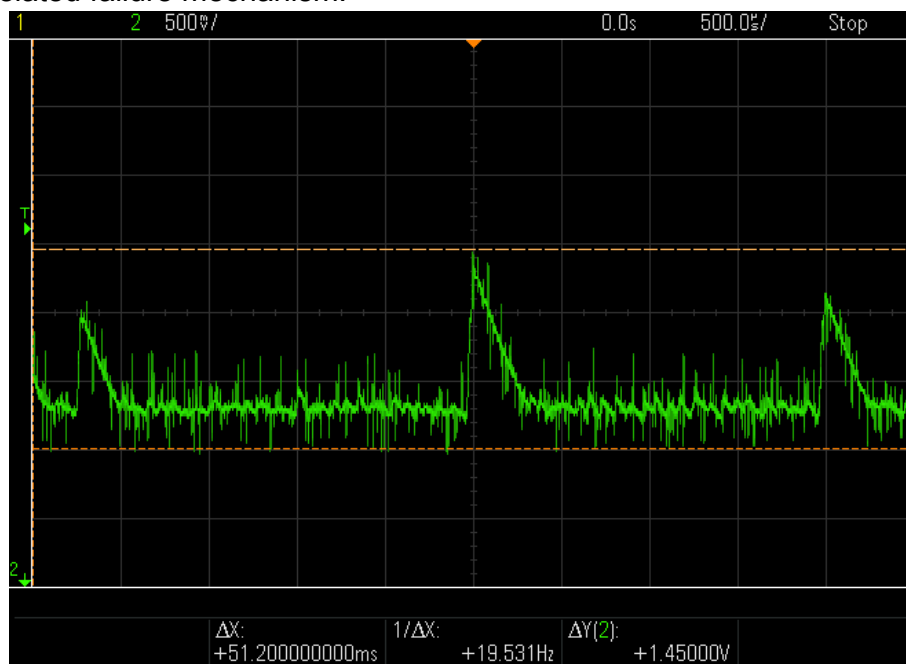


Figure 3.4.1.1. Expected Vc Waveform During Correct Operation

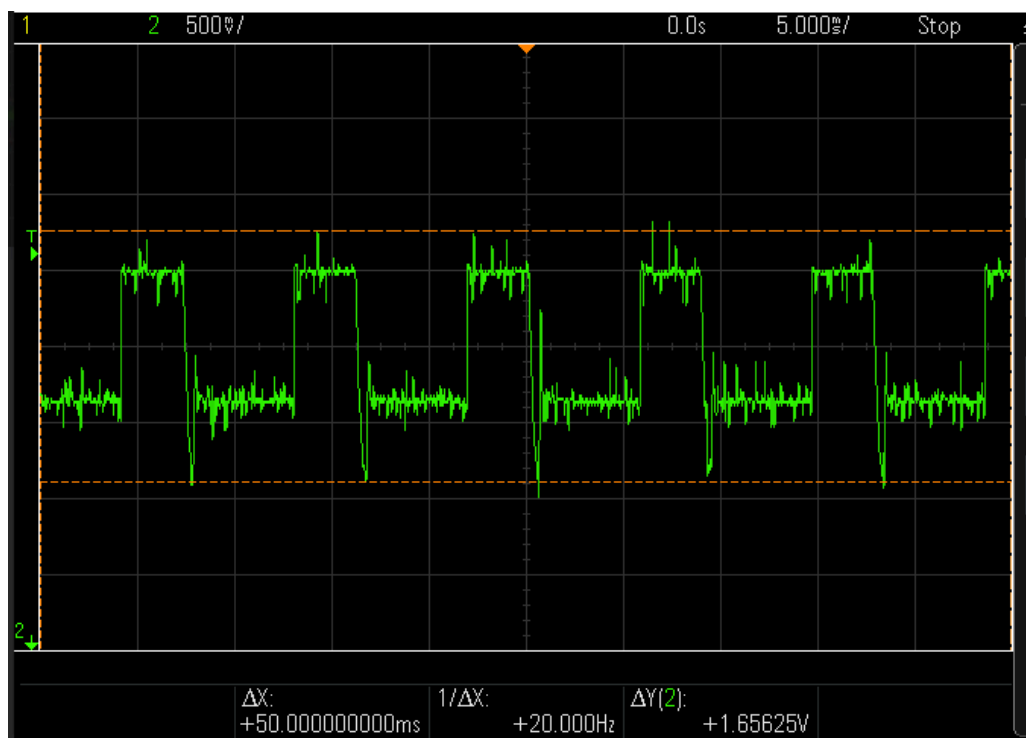


Figure 3.4.1.2. Vc Waveform at Failure

In this PCB revision, the SGND, VC, FB, and VCC/MPPC/SVCC decoupling networks were tied directly into the global ground polygon that also carries the high-di/dt switching currents of the power stage, preventing the LTC3119's analog control circuitry from having a quiet and stable ground reference. As output load increases, the corresponding rise in switching-current amplitude creates significant ground bounce across the shared plane, causing SGND and the internal VCC reference to shift relative to true ground. This noise couples into the VCC/VC node, contaminating the control voltage with switching artifacts and destabilizing the error amplifier. The distorted VCC waveform observed during failure directly confirms this behavior. In addition to the grounding issue, the LTC3119 IC was overheated during installation, and this likely weakened the internal bias regulator or error-amplifier structures, reducing their noise immunity and narrowing the device's usable control-voltage margin. While the grounding flaw alone can explain degraded current capability, the combination of ground noise and latent thermal damage makes the control loop far more susceptible to collapse as load increases, ultimately limiting the converter to approximately 0.93 A well below its intended operating range.

The failure of the 12 V converter at approximately 0.93 A is therefore attributed to the combined effects of improper grounding layout and thermal overstress of the LTC3119 IC during assembly. The shared analog/power ground return allowed switching noise to corrupt the VCC/VC control node, while the overheated IC exhibited reduced tolerance to this disturbance, causing the control loop to collapse and forcing the device into its internal protection mode at elevated load. A future hardware revision that provides a properly isolated SGND region, a short and direct return path for the VCC/MPPC/SVCC decoupling capacitor, and improved assembly temperature control is expected to eliminate this failure mode and restore the converter's full rated output-current capability.

3.5. ESP32 and TFT Touchscreen

3.5.1. Operation

The touchscreen and relay-control subsystem forms the primary user interface and digital switching layer of the Physical Controls System. It is implemented using two ESP32-S3 microcontrollers that divide responsibility between graphical user interaction and relay actuation. The subsystem is powered by the regulated 3.3 V domain generated from the main supply, which supports the logic-level electronics, while the 12 V rail provides the energy required for relay coil activation through isolated transistor drivers.

The first ESP32-S3 operates as the human-machine interface (HMI) and is responsible for rendering the graphical control panel on a 2.8-inch ILI9341 TFT display. An FT6206 capacitive touch controller detects user interaction with the displayed buttons. Each button corresponds to one of the five lighting channels in the system and is dynamically color-coded—green when the associated relay is active and red when inactive. This microcontroller translates user presses into structured UART messages, such as “R2:ON” or “R4:OFF,” which are transmitted to the relay-control ESP32 for execution.

The second ESP32-S3 performs all relay-driving functions. Upon receiving a command from the HMI controller, it decodes the message, activates or deactivates the appropriate transistor stage, and energizes the corresponding 12 V relay coil. This arrangement electrically isolates the logic circuitry from the higher-voltage load domain while ensuring robust and deterministic switching behavior. After completing an action, the relay microcontroller returns an acknowledgment message to the HMI so that the graphical interface remains synchronized with the actual system state at all times.

To enhance usability and reduce energy consumption, the subsystem incorporates automatic backlight management. The HMI includes a settings interface that allows manual adjustment of screen brightness, and inactivity triggers an automatic reduction or shutoff of the TFT backlight. Touch input immediately restores illumination, ensuring that the interface remains responsive even during extended idle periods. Both ESP32-S3 units include dedicated power-indication LEDs, providing immediate visual confirmation of subsystem power status during operation and testing.

Together, the UART communication link, touchscreen interface, relay-control logic, and coordinated power-management features create a responsive, low-power, and user-friendly subsystem capable of reliably controlling lighting loads across the full 10.8–14.2 V input range. The following subsection presents the detailed validation results demonstrating the performance and correctness of this architecture.

3.5.2. Validation

A comprehensive validation sequence was performed to confirm the functional correctness, reliability, electrical integrity, and command-response behavior of the Touchscreen and Relay-Control Subsystem. Validation included initial power-on behavior, touchscreen

mapping confirmation, long-duration reliability testing, and command-trigger validation under repeated use. All tests were conducted using direct bench instrumentation

3.5.2.1. Power-On Behavior and Initialization

When the subsystem was energized, both microcontrollers successfully booted without brownout or instability, and the TFT display initialized immediately. The capacitive touch controller and backlight system were detected and configured correctly at startup. These results confirm proper wiring, sufficient current delivery from the 3.3 V converter, and correct boot configuration for both ESP32-S3 modules.

3.5.2.2. Touchscreen Mapping and Interface Validation

The touch panel was exercised across the full display area to verify that the FT6206 controller correctly reported coordinate ranges and that all interactive elements aligned with their graphical regions.

During calibration, the screen was touched near each corner to establish the usable coordinate bounds. The measured coordinates were:

- **Top-Left:** (X = 10, Y = 320)
- **Bottom-Left:** (X = 237, Y = 318)
- **Top-Right:** (X = 20, Y = 2)
- **Bottom-Right:** (X = 237, Y = 1)

These readings confirmed that the touch controller reported stable, repeatable values and that the usable coordinate space spanned the full physical screen area, with minor offsets attributable to finger size and bezel margin. These boundary readings were then used to map the buttons precisely onto the display so that the graphical elements matched the physical touch detection regions.

After calibration was applied, all on-screen control buttons responded accurately within their intended areas, with no false triggers, edge misalignments, or missed presses observed during evaluation.

3.5.2.3. System Runtime Stability

A continuous runtime test was conducted over a two-hour period. After approximately 45 seconds of inactivity, the automatic backlight-sleep feature engaged as expected, reducing display power while keeping the system responsive. Throughout the two-hour session, the HMI and relay-control ESP32 modules remained fully operational, with no resets, freezes, communication failures, or degraded touch response. User interaction immediately restored the backlight, confirming proper performance of the sleep/wake logic.

3.5.2.4. Relay Trigger

The command-response behavior between the two ESP32-S3 modules was tested by actuating each of the five relay channels 50 times. This produced a total of 250 individual relay operations. Every press resulted in the correct UART command being transmitted to the relay controller, followed by successful relay activation and acknowledgment. Voltage measurements confirmed that each relay correctly delivered 12 V to its output terminals on every activation cycle. No missed triggers, delayed responses, or desynchronization events were observed during testing. The consistency of these results demonstrates that the UART protocol, relay drivers, transistor interfaces, and synchronization logic operate reliably under repeated use.

3.6. Subsystem Conclusion

The Physical Controls Subsystem integrates the low-voltage power stages and digital interface components necessary to operate the lighting and user-interaction elements of the Smart Solar Residential Lighting and Security System. Across multiple validation procedures, the subsystem demonstrated strong performance in the majority of its functional domains, with reliable conversion efficiency, stable user-interface behavior, and consistent relay-control operation.

The 3.3 V buck converter met its regulation and efficiency expectations, maintaining output stability across the full 10.8-14.2 V input range and up to 2 A of load. Line and load regulation remained within acceptable bounds, ripple measurements were low, and percent-error results confirmed accurate voltage control throughout all tested operating points. This converter is therefore verified as suitable for supplying all logic-level electronics within the subsystem.

The 12 V buck-boost converter exhibited stable voltage regulation and good efficiency up to approximately 0.8-0.85 A, but a hardware failure prevented testing within the intended current range of 2 A nominal and 3 A peak. Ripple levels and percent-error measurements reflect this limitation and represent behavior under degraded conditions rather than the expected performance of a properly functioning design. Despite this, the partial results provide insight into the converter's pre-failure behavior and confirm that the control architecture, layout, and test protocol remain valid for future hardware revisions.

The ESP32 touchscreen and relay-control architecture performed reliably across all validation criteria. The human-machine interface consistently detected touch inputs, mapped coordinates accurately, and communicated deterministically with the relay controller through UART. Long-duration runtime testing confirmed stable operation with no resets or lockups, and all 250 relay-activation cycles succeeded without error. The subsystem's responsiveness, synchronization, and low-power backlight management further demonstrated its suitability as the central user-interaction layer for the system.

Overall, the Physical Controls Subsystem achieved its functional objectives, with the exception of the identified fault in the 12 V converter hardware. All other components met or exceeded design expectations, validating the subsystem's readiness for integration with the solar-charging, sensor, and mobile-application subsystems. The results confirm that the architecture, interface definitions, and control logic provide a strong foundation for reliable system performance in the final Smart Solar Residential Lighting and Security platform.

4. AI Security Camera Subsystem

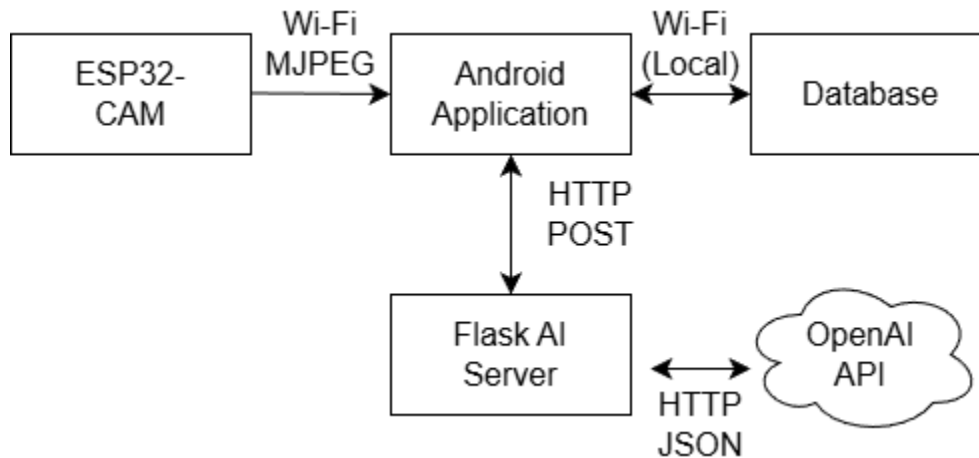
4.1. Subsystem Introduction

The AI enhanced Wireless Camera Subsystem is responsible for acquiring visual data from the ESP32-CAM module, transmitting the image stream to the android application, and enabling automated scene analysis through a cloud based AI model. This subsystem exists to provide real time visual monitoring, remote access, and intelligent feedback without requiring wired infrastructure or a local compute board. This provides easy accessibility for any user looking to implement the solar home lighting project. The subsystems integrate three major components: ESP32-CAM, Android Application, and Flask AI backend. The ESP32-CAM captures frames using an OV2640 sensor and exposes them over an MJPEG HTTP stream. The android application receives, decodes, displays, and stores frames while giving user remote control over lighting and analysis capabilities. The Flask AI backend receives short bursts of images from the app and uses OpenAI's model for automated, real time monitoring for security purposes. Together, these components form the imaging and intelligence backbone of the entire project.

4.2. Subsystem Overview

A block diagram of the subsystem is shown below.

Figure 4.2: Functional Block Diagram of AI Security Camera Subsystem



The primary focus of the AI Camera Subsystem was ensuring that all components would communicate properly and in a timely manner over local Wi-Fi and ensuring easy accessibility within the user application. Each component was successfully integrated over local Wi-Fi and displayed in a reliable manner as validated by the validation tests below.

The frame rate stability, connection verification, and AI Analysis latency were validated after the subsystem was mostly completed, and preformed within expectation laid out in the validation plan

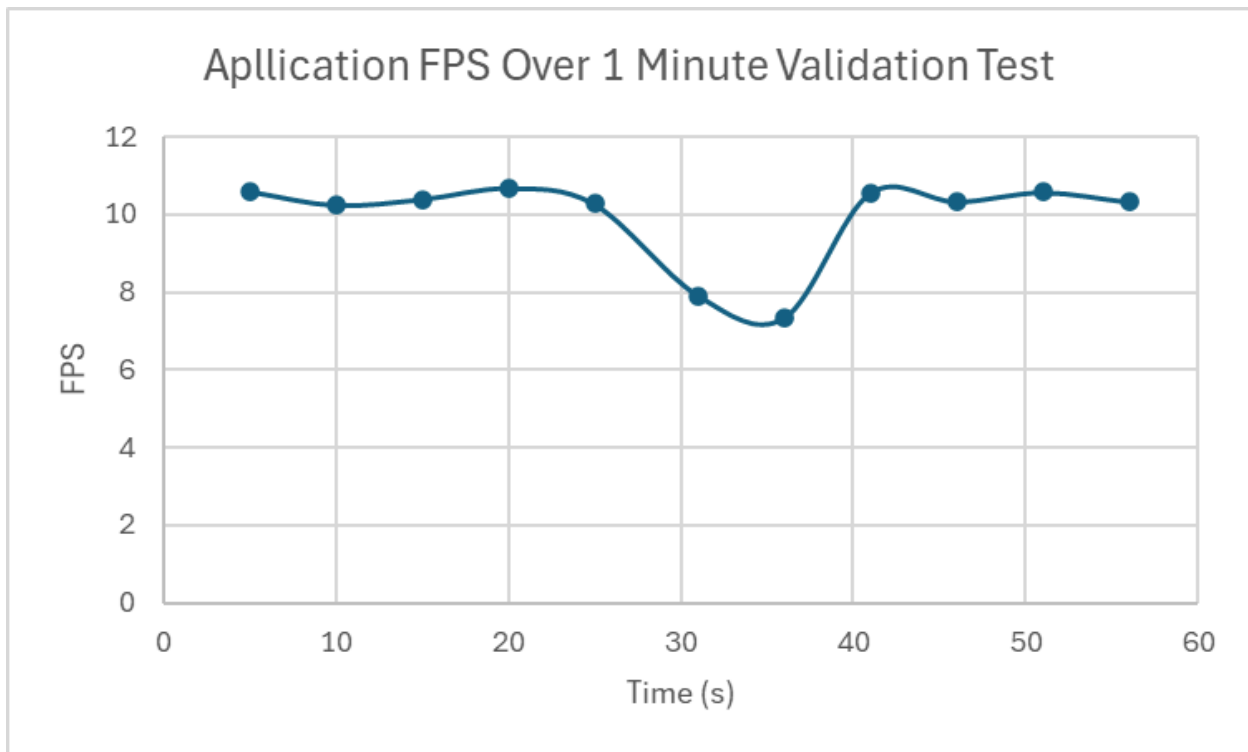
4.3. Subsystem Validation

To validate the reliability of the camera subsystem, quantitative tests were conducted on stream performance, AI pipeline latency, and network stability. Testing focused on real world conditions that the system would face during normal operation.

4.3.1. Frame Rate (FPS) Stability Test

A 60 second FPS sampling test was implemented inside the camera fragment portion of the application. The application logged average FPS every 5 seconds producing a time series of fram throughout under normal Wi-Fi conditions

Figure 4.3.1: Application FPS Over 1 Minute Validation Test



The purpose of this validation test was to evaluate the consistency of the MJPEG decoding pipeline and verify that the ESP32-CAM can maintain real time performance while streaming locally within the application. A table detailing exact frames is shown below.

Table 4.3.1: Application FPS Over 1 Minute Validation Test Data

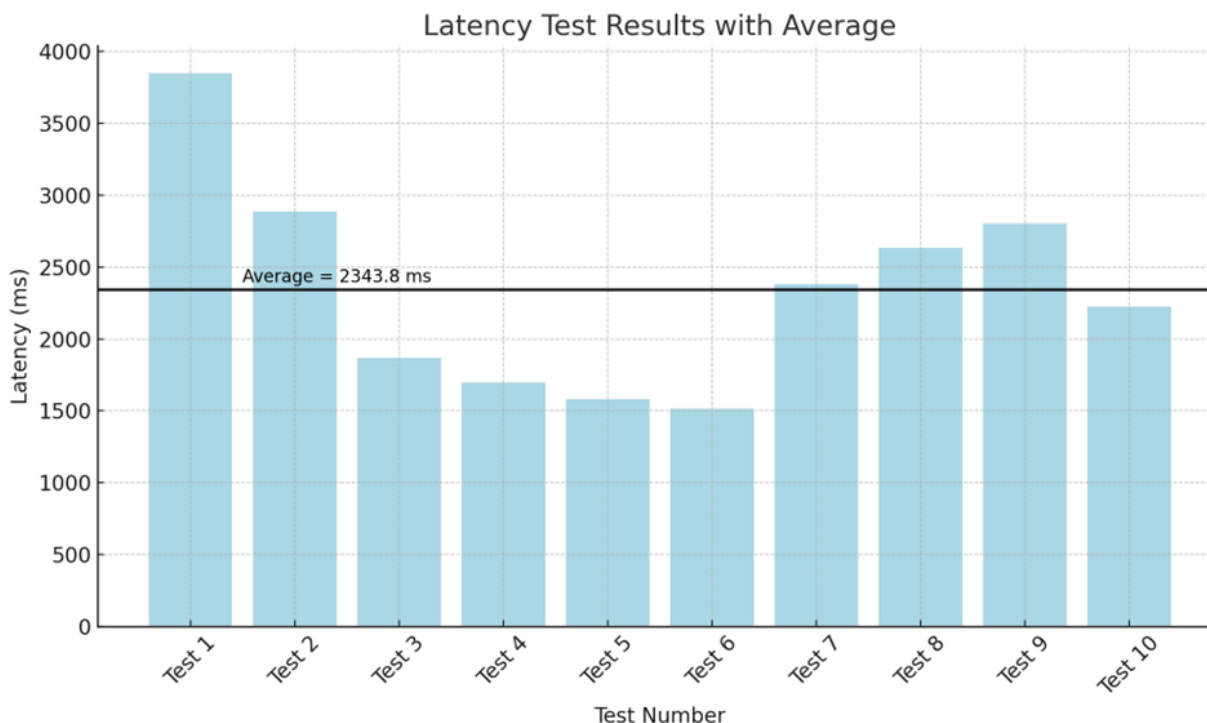
Time (sec)	FPS
5	10.61
10	10.25
15	10.4
20	10.69
25	10.28
31	7.91
36	7.32
41	10.56
46	10.34
51	10.58
56	10.33
Average FPS	9.93

The subsystem demonstrated stable performance with minimal drift, confirming a reliable streaming pipeline.

4.3.2. AI Latency Test

To quantify AI processing responsiveness, the app recorded the elapsed time from image upload to summary reception. This was done Time stamping each process from the start of the AI analysis, adding the total and running the test 10 times.

Figure 4.3.2: Latency Test Results with Average Graph



There was an average of 2.3s Latency for the whole system to return back to the application and display, which is well under the projected hypothesis of less than 5 seconds. An exact measurement of the test data is shown below.

Table 4.3.2: Latency Test Results Data

Test #	Time (ms)
1	3849
2	2885
3	1868
4	1696
5	1582
6	1513
7	2383
8	2633

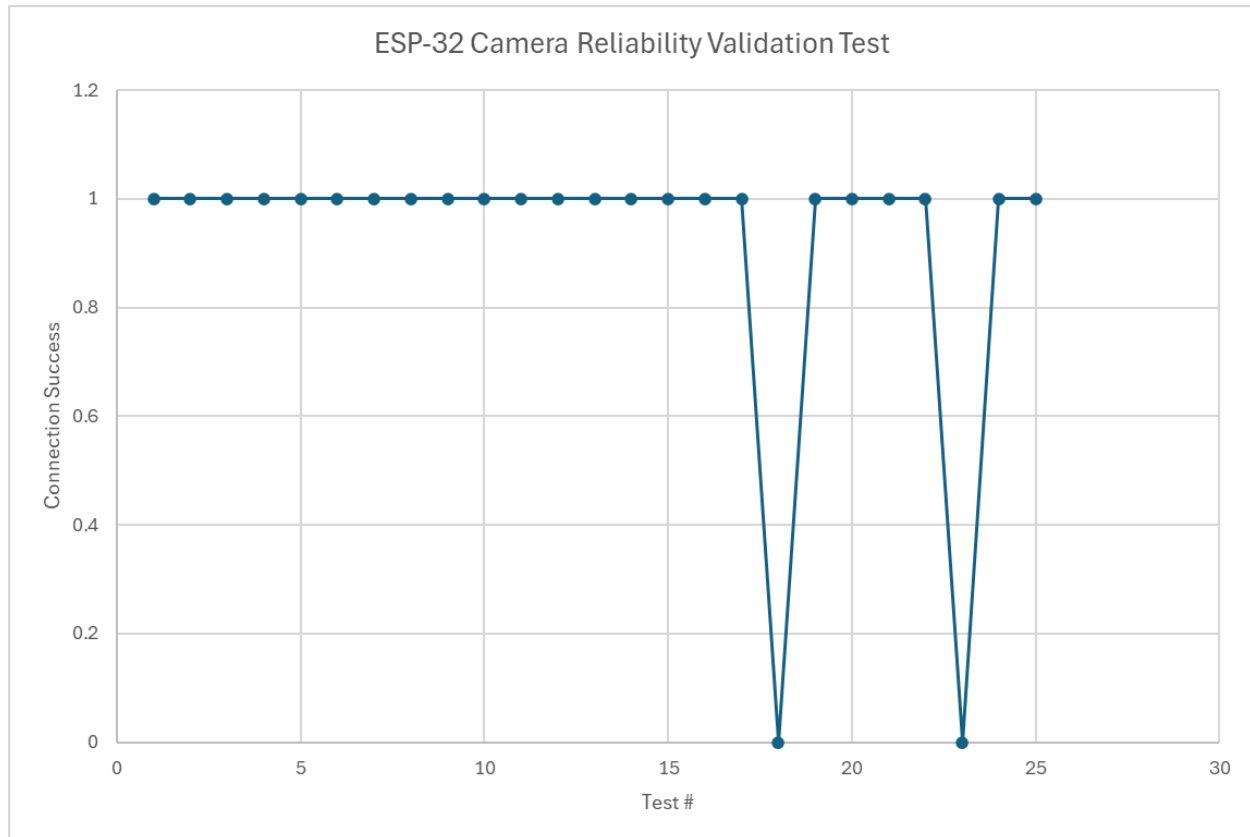
9	2803
10	2226
Average	2343.8

This conclusively validates that the AI backend is fast enough for real time environmental feedback.

4.3.3. Connection Robustness Test

This validation test was performed to check the connectivity from the Application to the ESP-32 Camera over 25 tries to ensure solid performance and communication between the two components

Figure 4.3.3: Camera Reliability Validation Test Visual Graph



The camera connected successfully 23 times with 2 failures due to connection timeout. This gives a 92% reliability within the camera which is above projected standards of 92% conclusively validating the robustness of the connection to the camera and application.

4.4. Subsystem Conclusion

Each part of the subsystem was shown to communicate and work successfully at a rate higher than projected initially. The Camera AI subsystem effectively streams MJPEG streams over local Wi-Fi, streams within the application, and streams and stores data at a reliable rate.

5. MCU, Sensors, and Database Subsystem

5.1 Subsystem Introduction

The MCU, Sensors, and Database subsystem serves as the central control layer of the Solar Home Lighting System, coordinating real-time sensing, decision-making, and two-way communication across the entire system. This subsystem interprets data from the charge controller as well as sensors including daylight levels and motion activity and uses these inputs to execute all system logic, such as determining night conditions, activating the porch light, and generating daily performance metrics.

In addition to sensor processing, the MCU maintains continuous communication with the cloud database, enabling both data upload and remote command retrieval. This bidirectional link allows the user to monitor system status and adjust configuration parameters through the application, which the MCU interprets and applies in real time. Through this architecture, the subsystem ensures synchronized behavior between the physical hardware and the user-facing app.

Through its combined sensing, control, and communication capabilities, the MCU, Sensors, and Database subsystem forms the decision-making backbone of the Solar Home Lighting System, ensuring coordinated and reliable operation under both real and simulated conditions.

5.2 Microcontroller Unit (MCU)

5.2.1 Operation

The Microcontroller Unit (ESP32) serves as the primary controller for the Solar Home Lighting System. The MCU periodically polls the charge controller to retrieve processed telemetry, including battery status, panel status, load behavior, and state of charge (SOC). In the current design, the MCU is configured to request this telemetry at an interval of 30 seconds, which is sufficient to capture system trends without overloading the communication link or database.

In addition to receiving telemetry from the charge controller, the MCU directly reads the daylight (LDR) and motion (PIR) sensors. The LDR is used to determine whether the system is in Day Mode or Night Mode, while the PIR sensor is used to detect motion events near the light. When the LDR indicates low light and the system enters Night Mode, any

valid motion detection triggers the MCU to energize the porch light relay for a configured duration and log the event.

The MCU takes the telemetry with local sensor readings to turn data into telemetry packets and daily performance metrics and uploads them to the cloud database at scheduled intervals. Over the same link, the MCU also retrieves configuration parameters and control flags from the user-facing application, allowing remote adjustment of thresholds, timers, and operating modes. In addition the MCU is responsible for controlling the relay logic balancing the response between the physical controls and application controls.

5.2.2 Validation

Validation of the Microcontroller Unit (MCU) focused on confirming correct telemetry polling, sensor-driven logic execution, relay activation, and database communication.

Table 5.2: MCU Subsystem Validation Summary

Validation Item	Procedure	Expected Result	Result
Charge-Controller Polling	Observe MCU requests at 30-second intervals	Telemetry received every cycle with no dropouts	Pass
Day/Night Detection State	Cover/uncover LDR	System switches between Day and Night modes	Pass
PIR Motion Detection	Trigger PIR during Night Mode	Motion event logged and relay activated	Pass
Relay Output	Measure relay ON duration	Relay deactivates when signaled	Pass
Database Upload	Inspect Firebase entries	Telemetry uploaded with correct timestamps	Pass
Remote Commands	Change app-control flags	MCU updates behavior after small latency	Pass
Simulation Mode	Generate bench telemetry	Simulated values appear in database as real data	Pass
Connectivity Loss Handling	Disable Wi-Fi during upload	MCU retries and uploads once reconnected	Pass

The MCU successfully polled the charge controller at the intended 30-second interval, with all telemetry received and logged correctly. Day/night transitions were validated using controlled changes in the LDR sensor, and the system reliably entered Night Mode at low light levels. PIR testing confirmed immediate relay activation and accurate timeout behavior. Remote override commands issued through the application updated relay and system behavior in real time.

Simulation mode produced predictable SOC and panel-status values that matched expected system behavior, confirming the MCU's ability to operate without live solar hardware. Connectivity loss testing showed that the MCU safely buffered data and retransmitted it once Wi-Fi was restored.

Overall, the MCU demonstrated reliable performance across all operational scenarios, confirming proper logic execution, sensor integration, relay control, and database synchronization.

5.3 Sensors

5.3.1 Operation

The Sensor Interface Layer consists of two primary environmental sensors: the Passive Infrared (PIR) motion sensor and the digital ambient light sensor used for daylight detection. Together, these sensors provide the MCU with the necessary information to determine lighting conditions and activate the porch light based on motion and environmental brightness.

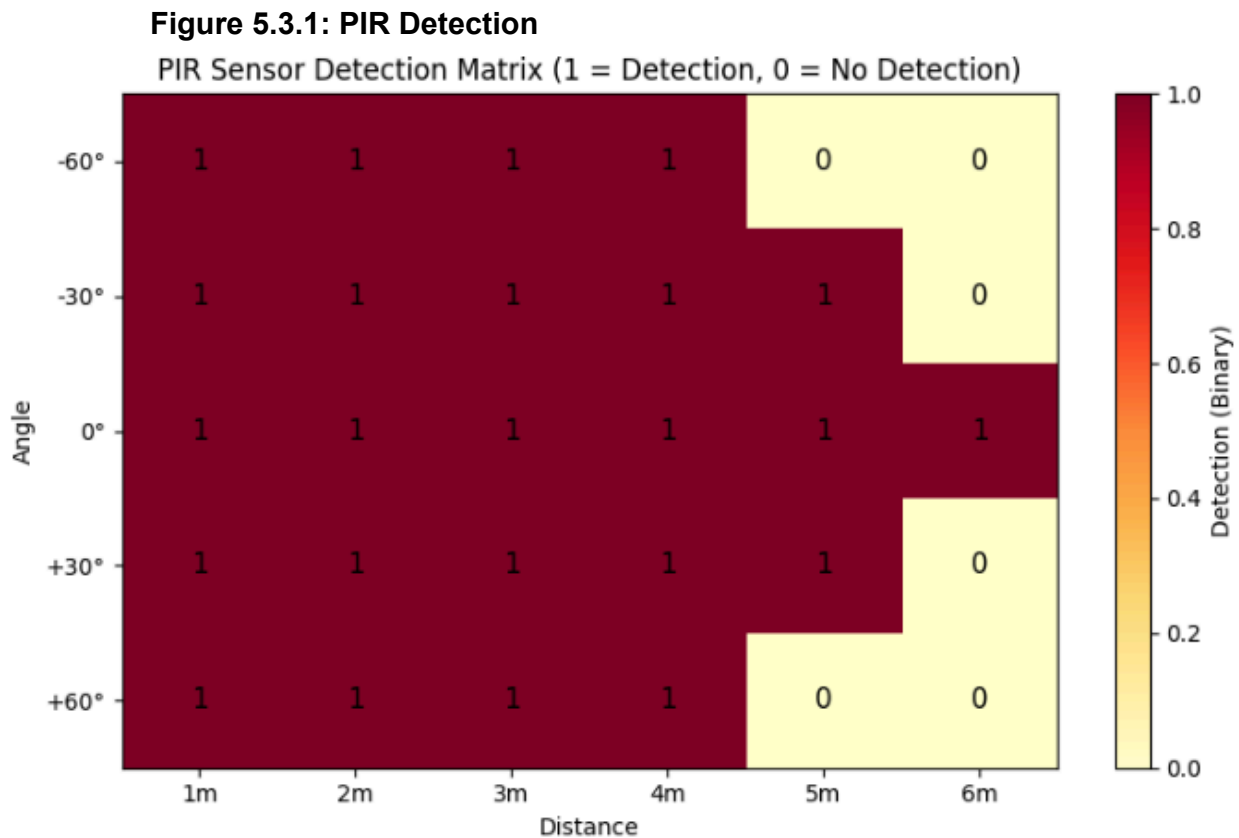
The PIR sensor detects infrared energy changes caused by human movement. When motion is detected, the sensor outputs a HIGH digital signal to the MCU. During Night Mode, this immediately triggers the security-light relay for the configured duration. The PIR sensor offers a practical detection range of approximately 3–5 meters and a horizontal field of view of roughly $\pm 60^\circ$, which is consistent with typical PIR hardware. This allows the system to reliably capture motion events in front-facing areas such as sidewalks or entry paths.

Instead of a traditional analog LDR, the system uses a digital ambient light sensor, which provides a raw brightness value through its digital output interface. The MCU periodically reads this brightness value and compares it to a predefined threshold to determine day and night conditions. When the sensor reading falls below the threshold, the MCU sets the internal state to dark=yes and transitions the system into Night Mode. When the reading rises above the threshold, the MCU returns to Day Mode (dark=no). This digital sensor provides stable and noise-resistant measurements, enabling consistent and reliable day/night classification for the lighting system.

5.3.2 Validation

Validation of the Sensor Interface Layer focused on verifying reliable PIR motion detection across practical angles and distances, as well as ensuring accurate day/night classification using the digital ambient light sensor. Both sensors were evaluated under controlled test conditions that reflect expected real-world operation.

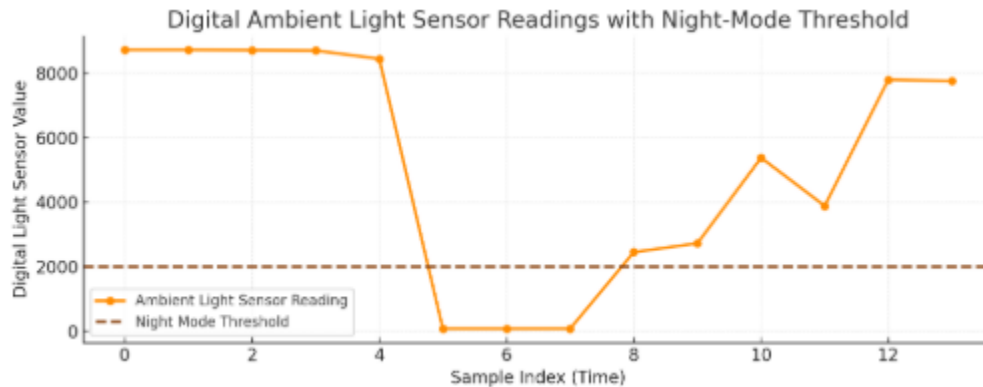
The PIR sensor was tested by recording binary detection results at multiple distances (1–6 meters) and angles (-60° to $+60^\circ$). Each test position was marked as either detected (1) or not detected (0). The resulting detection matrix was visualized to illustrate the sensor's effective coverage.



As shown in Figure 5.3 the PIR sensor reliably detected motion within 3–4 meters across $\pm 60^\circ$, with diminishing sensitivity beyond 5–6 meters and at extreme angles. This behavior is consistent and confirms the sensor's suitability for night-time motion-based lighting activation.

The digital light sensor was validated by measuring brightness readings during controlled lighting changes. The sensor output was logged while transitioning from bright indoor conditions to partial shade and complete darkness. A threshold value (≈ 2000 units) was used to determine when the system should enter Night Mode.

Figure 5.3.2: Digital Light Sensor Readings



The digital ambient light sensor outputs high values (approximately 8400–8800) under bright conditions. At sample index 5, the sensor is covered or exposed to low light, causing the reading to drop sharply below 100. This is well below the night-mode threshold of ~ 2000 units, prompting the MCU to classify the environment as dark (dark=yes). As light increases again across samples 8–13, the readings rise above the threshold and the system correctly returns to Day Mode (dark=no). This behavior confirms that the sensor accurately distinguishes between bright and dark conditions and the MCU interprets the measurements to control the porch light.

5.4 Database

5.4.1 Operation

The Database Subsystem serves as the communication bridge between the Solar Home Lighting System and the user-facing application. The subsystem is implemented using a real-time cloud database, which stores system telemetry, daily performance metrics, configuration parameters, and command flags. The MCU communicates with the database over Wi-Fi, enabling synchronization between the physical hardware and the mobile application.

The MCU periodically uploads charge-controller telemetry (battery status, SOC trend, panel state) along with local sensor readings and event logs. These values are timestamped and stored as structured entries within the database. In addition to raw telemetry, the MCU generates daily summary metrics such as energy-related trends which are also uploaded for visualization in the user interface.

Communication is bidirectional with the database receiving commands and set thresholds from the database. The MCU regularly looks at the database to see if the values have been updated to change these fields real time in the system

5.4.2 Validation

Validation of the Database Subsystem focused on confirming if the data uploaded reliably with consistent timestamping, successful retrieval of remote command and proper system behavior during connection loss. Validation was performed using test bench telemetry data, intentional command changes from the app and simulated wifi interruptions.

Figure 5.4: Database Telemetry



The validation of telemetry uploads and daily summary generation was validated by inspecting entries in the cloud database. As shown in figure 5.4 each daily record is timestamped and contains the relevant performance metric computed by the MCU. These results verify that the database subsystem stores structured, complete, and timely metrics during normal operation.

5.5 Conclusion

The MCU, Sensor and Database subsystem across all validation procedures demonstrated reliable sensing, decision making and data management. The PIR sensor consistently detected motion within its expected range and field of view, while the digital ambient light sensor provided stable day/night classification.

The MCU correctly integrated sensor inputs with charge-controller telemetry, executed system logic in real time, and validated proper relay activation during Night Mode. The database subsystem consistently stored data received and logged events correctly while communicating with the MCU. Overall, the subsystem met all functional requirements for real-time monitoring. These results confirm that the MCU, Sensor and Database subsystem is synchronized and works reliably, automating the lighting behavior and tracking system performance for the Solar Home Lighting System.